Counterfactual Invariance via Conditional Mean Embeddings: Robustness to Domain Generalisation for Tabular Data

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11 June 2026

Submitted in partial fulfillment of a structured master's degree at AIMS South Africa



Abstract

Machine learning models learn distributions from finite observational data without knowledge of underlying causal relations, risking exploitation of spurious correlations rather than stable causal structures. The methods used in this project may be used to solve three types of problem on large tabular datasets under a unified framework of counterfactual invariance: spurious correlation reduction, domain generalisation, and individual fairness. We enforce invariance of model predictions to a designated counterfactual feature using the Hilbert-Schmidt Conditional Independence Criterion (HSCIC). Our contribution operationalises this on tabular classification via a warm-start strategy that decouples task learning from invariance enforcement. This approach is computationally efficient, skipping kernel costs during warm-up while achieving counterfactual invariance comparable to that of joint training from a random initialisation. We confirm in a classification task that HSCIC and the oracle counterfactual invariance certificate, Variance of Counterfactuals (VCF), decrease together as the trade-off parameter increases. Our experiments demonstrate that invariance can be learned without a catastrophic drop in performance. Furthermore, the Counterfactual Invariance Prediction approach successfully outperforms feature deletion in synthetic and real-world setting. Crucially, we identify a breakdown at high trade-off parameter values where the HSCIC continues to decrease while the oracle VCF increases; this discordance happens alongside a notable drop in performance. Finally, we confirm that learned invariance transfers out-of-domain without retraining; however, predictive robustness transfer is fully realised only in synthetic settings since we can control the intensity of the out-of-domain shift on synthetic data generated from known structural causal model.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Michael Fitiavana RANDRIAMIFIDY, 11 June 2026

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1. Introduction

1.1 Context

Machine learning models are trained on observational data and use correlation to make predictions. A model may attribute importance to a feature that domain knowledge considers irrelevant. In this situation, a practitioner would expect the prediction to stay the same if the value of that feature were changed. When this expectation is violated, the feature is said to carry a *spurious correlation* (Veitch et al., 2021).

One common cause of spurious correlations is a *confounder*: a variable that affects both the input features and the target. Because the confounder drives both variables simultaneously, the model interprets the resulting association as the input causing the output, even though no direct causal link exists (Pearl, 2009). If a domain shift is caused by an intervention on such a variable affected by the confounder, the statistical pattern the model has learnt during training breaks down, degrading out-of-domain performance. It is therefore preferable to learn a model that is invariant to changes in those features that have shifted.

Fairness provides a concrete instance of this problem. In *group fairness* we ask whether the model performs equally across groups defined by a sensitive attribute A. We measure this group fairness using Equalised Odds (Hardt et al., 2016). In *individual fairness* we ask a stronger question: would the decision remain the same if the same individual had a different value of A? This counterfactual scenario is never directly observed. Kusner et al. (2017) formalise it as *counterfactual fairness*, and we use this notion in a more general setting: we ask what the distribution of the output $\hat{Y}$ would be if the features A took different values, as defined by Quinzan et al. (2024).

A naive approach to enforcing this invariance is to change the value of A while keeping all other features fixed. However, because features are generated jointly by the underlying causal model, this can produce input combinations that are unlikely or incoherent (Lara et al., 2024). Feeding such unrealistic inputs to the model either corrupts the training signal or causes the model to ignore those regions entirely. Approaches such as counterfactual fairness (Kusner et al., 2017) explicitly penalise dependence of model outputs on counterfactual variables. In this project, we focus on learning a model to predict under a counterfactual invariance constraint formulated with Kernel based method.

1.2 Research Questions and Contributions

The central question of this thesis is: for tabular data, can we train a machine learning model using a gradient-based optimiser to satisfy counterfactual invariance with respect to a set of interventional variables A and meet the following criterion: the predictions do not change under interventions on feature A while retaining A in the input and without a catastrophic loss of predictive performance. This question matters because simply removing A from the input, the standard response to spurious correlations, is insufficient. Indeed, proxy features correlated with A remain in the data and allow the model to recover the same dependence indirectly (Kusner et al., 2017). Moreover, deletion discards the informative signal that A may carry, leading to degraded in-domain and out-of-domain performance (Nastl and Hardt, 2024). We argue that the right objective is not to prevent the model from seeing A, but to prevent it from relying on A to make a prediction. We use a Kernel-Based Approach and the Hilbert-Schmidt Conditional Independence Criterion (HSCIC) to build a condition to guarantee Counterfactual Invariance (Park and Muandet, 2020; Quinzan et al., 2024).

This thesis addresses the following specific questions.

Does minimising HSCIC certify counterfactual invariance? In the synthetic setting, where the true SCM is known and counterfactuals can be generated exactly, we compare the HSCIC estimate against the Variance of Counterfactuals $\mathrm{VCF}(\hat{Y})$ of Eq. (3.3.2). The hypothesis is that the statistical estimand HSCIC and the theoretical quantity VCF decrease together when we increase the trade-off parameter.

How does enforcing invariance affect predictive performance? A stronger invariance constraint limits the model's reliance on the counterfactual features, whilst preserving competitive predictive performance.

Is the Counterfactual Invariance Prediction (CIP) approach better than feature deletion? We compare CIP against a baseline that removes A from the input entirely. The hypothesis is that CIP, by retaining A and penalising its use, achieves lower HSCIC than deletion while matching or exceeding deletion on in-domain AUC-ROC, because it does not discard the causal signal that A carries.

Does counterfactual invariance in A still hold under domain shift? We evaluate whether the counterfactual invariance holds out-of-domain without retraining.

Does fine-tuning a pre-trained model to be counterfactually invariant achieve the same result as a model randomly initialised and trained with the invariance constraint from scratch? We compare the two CIP strategies across the full β sweep. The hypothesis is that Strategy 2 (warm-start) reaches a better solution at high β because the task loss has already converged before the invariance constraint is applied, giving the model a more informative initialisation.

1.3 Outline

The chapters break down as follows.

Chapter 2 provides the formal background required for the rest of the thesis. We introduce Directed Acyclic Graphs (DAGs) and Structural Causal Models (SCMs), characterise the roles played by nodes in a causal DAG. We then present the functional-analytic machinery used throughout: Reproducing Kernel Hilbert Spaces (RKHS), Kernel Mean Embeddings (KME), Maximum Mean Discrepancy (MMD), and Conditional Mean Embeddings (CME).

Chapter 3 develops the framework for counterfactual reasoning and formalises the notion of counterfactual invariance. We derive an independence condition in the RKHS and define the counter-factually invariant predictor that forms the target of our learning framework.

Chapter 4 introduces the Counterfactual Invariance Prediction and details every aspect of the experimental design. We describe the synthetic data generating process, including node roles, structural equations, and the domain-shift mechanism. We then present the real-data classification task (ACS income classification) and the four experimental configurations: an unconstrained baseline on full features (Baseline 0), feature deletion (Baseline 1), joint CIP training from epoch 0 (Strategy 1), and a warm-start strategy in which the task loss is first minimised before the invariance constraint is applied (Strategy 2). The chapter concludes with details of the model architecture, optimiser, training loop, cross-validation protocol, randomness control, and evaluation metrics, including the Variance of Counterfactuals (VCF), HSCIC, AUC-ROC, and SHAP-based feature attribution.

Chapter 5 presents and discusses the results of both the synthetic and real-data experiments. On synthetic data we examine the trade-off between counterfactual invariance and predictive performance as a function of the regularisation coefficient β, and we compare the joint and two-phase training strategies.

On real data we investigate whether feature deletion successfully blocks proxy pathways, analyse the relationship between MMD and performance, and interpret the effect of β on model explanations via SHAP values.

Chapter 6 summarises the main findings, discusses the limitations of the approach and future work.

2. Background

2.1 Directed Acyclic Graph and Structural Causal Model

2.1.1 Definition (Directed Acyclic Graph)**.** Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed graph, where $\mathcal{V}$ denotes a finite set of vertices and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ a collection of directed edges. $\mathcal{G}$ is called a Directed Acyclic Graph (DAG) whenever it admits no directed cycle, i.e., there is no sequence of vertices $v_1, v_2, \ldots, v_k \in \mathcal{V}$ with $k \geq 2$ such that $(v_i, v_{i+1})_{i=1}^{k-1}$ and (v_k, v_1) are elements of the set of directed edges $\mathcal{E}$. We denote by $\mathrm{Pa}(v_i)$ the set of parents of the node v_i. We denote by $\mathrm{De}(v_i)$ the set of descendants of v_i; $\mathrm{Nd}(v_i) = \mathcal{V} \setminus (\mathrm{Pa}(v_i) \cup \mathrm{De}(v_i) \cup \{v_i\})$ denotes the set of non-descendants of v_i in $\mathcal{G}$. These sets may be empty. A path π between nodes A and C is any sequence of adjacent edges, ignoring edge directions.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, $\mathcal{V} = \{V_i\}_{i=1}^p$ be a set of endogenous (observable) random variables, $\mathcal{U} = \{U_1, \ldots, U_p\}$ a set of exogenous (background) variables, and $\mathbb{U}$ be the joint distribution of the exogenous variables. We also assume that the background variables are mutually independent. Throughout this document, the notation for a given random variable also denotes the associated node in the directed acyclic graph.

2.1.2 Definition (Structural Causal Model (Pearl, 2009))**.** Let $\mathcal{U}$, $\mathcal{V}$ be sets of variables with $p = |\mathcal{U}| = |\mathcal{V}|$, $U = (U_1, \ldots, U_p)$ where each $U_i \in \mathcal{U}$, $V = (V_1, \ldots, V_p)$ where each $V_i \in \mathcal{V}$, $\mathcal{F} = \{f_i\}_{i=1}^p$ a collection of functions, and $\mathbb{P}_U$ the joint distribution of U. A Structural Causal Model (SCM) is a tuple $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, \mathbb{P}_U)$, where $\mathcal{U}$ are the exogenous variables, $\mathcal{V}$ the endogenous variables, $\mathcal{F}$ the structural equations, and $\mathbb{P}_U$ the exogenous distribution. Each structural equation is given by

$$V_i \coloneqq f_i\Big(\mathrm{Pa}(V_i), U_i\Big), \qquad i = 1, \ldots, p;$$

$\mathrm{Pa}(V_i) \subseteq \mathcal{V} \setminus \{V_i\}$ denotes the parents of V_i (in causal terms, the set of all direct causes of V_i within $\mathcal{V}$), $U_i \in \mathcal{U}$ is the specific exogenous noise of V_i.

The SCM $\mathcal{M}$ uniquely determines a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $(V_i, V_j) \in \mathcal{E}$ if and only if $V_i \in \mathrm{Pa}(V_j)$ (i.e., V_j is a function of V_i in the structural equation). This graph is called the Causal Directed Acyclic Graph. We then have the following factorisation of the joint density function of $(V_1, \cdots, V_p)$, called the Bayesian Network Factorisation (Pearl, 2009, Sec. 3.2).

$$p(v_1, \cdots, v_p) = \prod_{i=1}^{p} p(v_i \mid \mathrm{Pa}(V_i)) \tag{2.1.1}$$

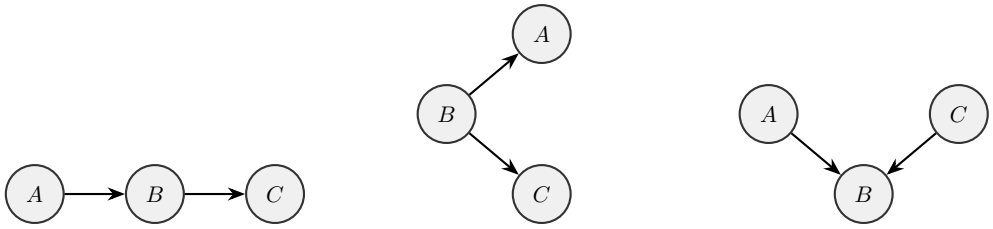
This follows from repeated application of Bayes' theorem and the local Markov condition (Pearl, 2009, Sec. 1.2.2). In terms of the graph, the Markov factorisation above encodes the property that each V_i is conditionally independent of its non-descendant nodes $\mathrm{Nd}(V_i) = \mathcal{V} \setminus (\mathrm{Pa}(V_i) \cup \mathrm{De}(V_i) \cup \{V_i\})$ given its parents $\mathrm{Pa}(V_i)$. We define this property as Local Markov Condition.

2.1.3 Definition (Local Markov Condition)**.** A probability $\mathbb{P}$ satisfies the local Markov condition with respect to the DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ if every variable $V \in \mathcal{V}$ is conditionally independent of all its non-descendants with respect to $\mathbb{P}$ given its parents. Denoting by W the random variable associated to $\mathrm{Nd}(V)$, in other words

$$V \perp\!\!\!\perp W \mid \mathrm{Pa}(V) \quad \text{for all } V \in \mathcal{V}.$$

2.2 Graph anatomy in a Causal DAG

Three fundamental path structures are the chain, fork, and collider, as depicted in Fig. 2.1 (Pearl, 2009). In a chain structure (a), given by $A \to B \to C$, node B transmits the causal effect from A to C. In a fork structure (b), represented by $A \leftarrow B \to C$, B acts as a common cause that confounds the relationship between A and C. Finally, in a collider structure (c), denoted by $A \to B \leftarrow C$, the node B naturally blocks the path between A and C.



(a) Chain (b) Fork (c) Collider

Figure 2.1: Three fundamental Directed Acyclic Graph (DAG) structures.

2.2.1 Definition (Causal Path between Two Sets of Nodes)**.** Let $\mathbf{X}$ and $\mathbf{Y}$ be two disjoint sets of nodes. A path $\pi = (V_0, V_1, \cdots, V_k)$ from $\mathbf{X}$ to $\mathbf{Y}$ is called a causal path whenever $V_0 \in \mathbf{X}$ and $V_k \in \mathbf{Y}$. Moreover, the path is called proper if V_0 is the only node in the path belonging to the set $\mathbf{X}$.

2.2.2 Definition (Blocked Path)**.** Let $\pi = (V_0, V_1, \ldots, V_k)$ be a path in $\mathcal{G}$ and let $\mathbf{Z} \subseteq \mathcal{V}$ with $V_0, V_k \notin \mathbf{Z}$. The path π is **blocked** by $\mathbf{Z}$ if there exists a node V_i $(1 \leq i \leq k-1)$ on π satisfying one of the following.

(i) V_i is a non-collider on π, i.e. $V_{i-1} \to V_i \to V_{i+1}$ or $V_{i-1} \leftarrow V_i \to V_{i+1}$, and $V_i \in \mathbf{Z}$;

(ii) V_i is a collider on π, i.e. $V_{i-1} \to V_i \leftarrow V_{i+1}$, and V_i has no descendant in $\mathbf{Z}$.

2.2.3 Definition (d-Separation)**.** Let $\mathbf{X}$, $\mathbf{Y}$, $\mathbf{Z}$ be mutually disjoint subsets of $\mathcal{V}$ in the DAG $\mathcal{G}$. $\mathbf{X}$ and $\mathbf{Y}$ are d-**separated** by $\mathbf{Z}$ in $\mathcal{G}$, written

$$(\mathbf{X} \perp_{\mathcal{G}} \mathbf{Y} \mid \mathbf{Z}),$$

if all paths linking any node $X \in \mathbf{X}$ to any node $Y \in \mathbf{Y}$ are blocked by $\mathbf{Z}$.

We can now define a relation between d-separation and the independence of two sets of random variables.

2.2.4 Definition (Global Markov Condition)**.** A probability $\mathbb{P}$ satisfies the global Markov condition with respect to the DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ if for any three mutually disjoint sets $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathcal{V}$,

$$\Big(\mathbf{X} \perp_{\mathcal{G}} \mathbf{Y} \mid \mathbf{Z}\Big) \implies X \perp\!\!\!\perp Y \mid Z.$$

Equivalently, any d-separation statement in $\mathcal{G}$ implies a conditional independence in $\mathbb{P}$.

2.2.5 Definition (Valid Adjustment Set)**.** Let $\mathbf{X}, \mathbf{Y}$ be disjoint node sets in the DAG. A set $\mathbf{S}$ is a valid adjustment set for $(\mathbf{X}, \mathbf{Y})$ if two conditions hold: (i) no element of $\mathbf{S}$ is a descendant in $\mathcal{G}_{\mathbf{X}}$ of any node $W \notin \mathbf{X}$ lying on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$; and (ii) $\mathbf{S}$ blocks all non-causal paths from $\mathbf{X}$ to $\mathbf{Y}$ in $\mathcal{G}$.

2.2.6 Definition (Intervention)**.** Let $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, \mathbb{P}_U)$, be a SCM where $\mathcal{F} = \{f_i\}_{i=1}^{p}$, $\mathbf{X} = \{X_i\}_{i=1}^{n} \subseteq \mathcal{V}$ a set of variables, $x = (x_i)_{i=1}^{n}$ where x_i is a realisation of X_i. An intervention $\mathrm{do}(\mathbf{X} = x)$

on $\mathbf{X}$ fixes the value of every $X_i \in \mathbf{X}$ to x_i, replacing its structural equation with the constant function $f_i \equiv x_i$. We denote by $\mathcal{M}_x = (\mathcal{U}, \mathcal{V}, \mathcal{F}_x, \mathbb{P}_U)$. The induced DAG $\mathcal{G}_{\bar{\mathbf{X}}}$ is obtained by removing all incoming edges into nodes in $\mathbf{X}$. The interventional distribution of $\mathbf{Y}$ under $\mathrm{do}(\mathbf{X} = x)$ is denoted by $\mathbb{P}_{Y|\mathrm{do}(\mathbf{X}=x)}$.

2.3 Reproducing Kernel Hilbert Space (RKHS)

We review kernels and reproducing kernel Hilbert spaces (RKHS). Let $\mathcal{X}$ be a nonempty set and let $\mathcal{H}$ be a Hilbert space of functions $f : \mathcal{X} \to \mathbb{R}$ with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and norm $\| \cdot \|_{\mathcal{H}}$ defined on $\mathcal{H}$.

2.3.1 Definition (Reproducing Kernel Hilbert Space)**.** The space $\mathcal{H}$ is called a reproducing kernel Hilbert space if there exists a symmetric function

$$k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$$

satisfying the two following properties.

(i) For every element $x \in \mathcal{X}$, the function defined by $k(x, \cdot) : x' \mapsto k(x, x')$ is an element of the Hilbert space $\mathcal{H}$.

(ii) (Reproducing property) For all $f \in \mathcal{H}$ and $x \in \mathcal{X}$,

$$f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}}.$$

The function k is called the reproducing kernel of $\mathcal{H}$ and the mapping from $\mathcal{X}$ to $\mathcal{H}$ defined by $\phi : x \mapsto k(x, \cdot)$ is called canonical feature map[1].

The RKHS is uniquely defined by the closure of the vector space generated by the set of all functions $\{k(x, \cdot) \mid x \in \mathcal{X}\}$. So

$$\mathcal{H} = \overline{\mathrm{span}\{k(x, \cdot) \mid x \in \mathcal{X}\}},$$

where the closure is taken with respect to $\| \cdot \|_{\mathcal{H}}$. Equivalently, any $f \in \mathcal{H}$ can be expressed as

$$f(\cdot) = \sum_{i=1}^{\infty} \alpha_i k(x_i, \cdot),$$

for some sequence $(x_i)_{i \geq 1} \subset \mathcal{X}$ and coefficients $(\alpha_i)_{i \geq 1} \subset \mathbb{R}$ depending on f and satisfying

$$\sum_{i,j=1}^{\infty} \alpha_i \alpha_j k(x_i, x_j) < \infty.$$

Indeed, every RKHS has a unique reproducing kernel, and conversely, for every positive definite kernel k on $\mathcal{X}$, there exists a unique RKHS $\mathcal{H}$ for which k is the reproducing kernel.

2.3.2 Definition (Positive Definite Kernel)**.** A symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called positive definite if for all sequences $(x_i)_{i=1}^n \subset \mathcal{X}$ and sequences of coefficients $(\alpha_i)_{i=1}^n \subset \mathbb{R}$

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j k(x_i, x_j) \geq 0, \text{ for all } n \in \mathbb{N}^*,$$

[1] Since the function k is symmetric, the function $k(x, .)$ and $k(., x)$ are equal. We can use these two functions interchangeably

In other words, the matrix $\mathbf{K}$ of $\mathbb{R}^{n\times n}$ defined by $\mathbf{K}_{ij} = k(x_i, x_j)$ for $1 \leq i, j \leq n$, is positive definite for all $n \in \mathbb{N}^*$. The most common kernels are

- Linear kernel: $k(x, x') = x^\top x'$,
- Polynomial kernel: $k(x, x') = (x^\top x' + c)^p$, with $p \in \mathbb{N}$, $c \geq 0$,
- Radial basis function kernel: $k(x, x') = \exp\left(-\frac{\|x-x'\|_2^2}{2\sigma^2}\right)$,
- Laplacian kernel: $k(x, x') = \exp\left(-\frac{\|x-x'\|_1}{\sigma}\right)$,

for all x, x' elements of the Euclidean space $\mathbb{R}^d$.

These last three kernels enable non-linear modelling through linear methods in high-dimensional feature spaces. In this project we use the Radial Basis Function kernel; σ is called the bandwidth.

2.4 Kernel Mean Embeddings (KME)

Kernel mean embeddings map a probability measure to an element of a Hilbert space, allowing distributional comparisons to be carried out via inner products (Smola et al., 2007). Let $\mathcal{H}_k$ be an RKHS with reproducing kernel k.

2.4.1 Definition (Kernel Mean Embedding (KME))**.** Let $\mathcal{P}$ be the set of all probability measures on the measurable space $(\mathcal{X}, \mathcal{B}_\mathcal{X})$ such that $\sup_{x\in\mathcal{X}} k(x, x) < \infty$. The kernel mean embedding is the mapping

$$\mu : \mathcal{P} \to \mathcal{H}_k, \quad \mathbb{P} \mapsto \mu_\mathbb{P} := \int_\mathcal{X} k(x, \cdot)\, \mathrm{d}\mathbb{P}(x), \tag{2.4.1}$$

The element $\mu_\mathbb{P} \in \mathcal{H}_k$ is also referred to as the kernel mean of $\mathbb{P}$. Moreover,

$$\mu_\mathbb{P}(x') = \int_\mathcal{X} k(x, x')\, \mathrm{d}\mathbb{P}(x)$$

is the Bochner integral of the feature map $k(x, .)$ with respect to $\mathbb{P}$.

In this document, for a random variable X having values in $\mathcal{X}$ and following the distribution $\mathbb{P}$, we equivalently write $\mu_X := \mu_\mathbb{P}$ and read it as "mean embedding of X".

The mean embedding $\mu_\mathbb{P}$ provides a representation of the distribution $\mathbb{P}$ in the Hilbert space $\mathcal{H}_k$. This representation is injective if and only if the kernel k is characteristic, meaning that

$$\mu_\mathbb{P} = \mu_\mathbb{Q} \quad \Longleftrightarrow \quad \mathbb{P} = \mathbb{Q}, \quad \text{for all } \mathbb{P}, \mathbb{Q} \in \mathcal{P}.$$

In this case, the embedding preserves all information about the distribution. The Gaussian and Laplace kernels are examples of characteristic kernels on $\mathbb{R}^d$.

We can estimate the mean embedding of a distribution $\mathbb{P}$ using Monte Carlo methods. Let $\{x_i\}_{i=1}^n$ be a sample of i.i.d. random variables $\{X_i\}_{i=1}^n$ having distribution $\mathbb{P}$. The empirical kernel mean embedding of $\mathbb{P}$ is

$$\hat{\mu}_\mathbb{P} := \frac{1}{n}\sum_{i=1}^n k(\cdot, x_i) \tag{2.4.2}$$

This estimator is unbiased.

2.5 Maximum Mean Discrepancy (MMD)

The Maximum Mean Discrepancy is an instance of an integral probability metric, a family of distances between distributions (Müller, 1997). Let $\mathcal{H}_k$ be an RKHS with reproducing kernel k.

2.5.1 Definition (Maximum Mean Discrepancy (MMD))**.** Let $\mathbb{P}, \mathbb{Q} \in \mathcal{P}$ be two probability measures on $(\mathcal{X}, \mathcal{B}_\mathcal{X})$. The MMD is defined as the Integral Probability Metric (IPM) over the class of functions in the unit ball of the RKHS $\mathcal{H}_k$.

$$\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) := \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left| \int_\mathcal{X} f(x) \, \mathrm{d}\mathbb{P}(x) - \int_\mathcal{X} f(x) \, \mathrm{d}\mathbb{Q}(x) \right|. \tag{2.5.1}$$

2.5.2 Proposition. $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k)$ is the norm of the difference between the mean embeddings of $\mathbb{P}$ and $\mathbb{Q}$

$$\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) := \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}.$$

Proof. The reproducing property gives $f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}$ for all $f \in \mathcal{H}_k$. Restricting to f with $\|f\|_{\mathcal{H}_k} \leq 1$ and applying linearity of the inner product,

$$\begin{aligned}
\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) &= \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left| \int_\mathcal{X} f(x) \, \mathrm{d}\mathbb{P}(x) - \int_\mathcal{X} f(x) \, \mathrm{d}\mathbb{Q}(x) \right| \\
&= \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left| \int_\mathcal{X} \langle f, k(x, .) \rangle \, \mathrm{d}\mathbb{P}(x) - \int_\mathcal{X} \langle f, k(x, .) \rangle \, \mathrm{d}\mathbb{Q}(x) \right| \\
&= \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left\langle f, \int_\mathcal{X} k(x, \cdot) \mathrm{d}\mathbb{P}(x) - \int_\mathcal{X} k(x, \cdot) \mathrm{d}\mathbb{Q}(x) \right\rangle \\
&= \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \langle f, \mu_\mathbb{P} - \mu_\mathbb{Q} \rangle .
\end{aligned}$$

If $\mu_\mathbb{P} = \mu_\mathbb{Q}$ the result holds trivially. Otherwise, by the Cauchy–Schwarz inequality, $\langle f, \mu_\mathbb{P} - \mu_\mathbb{Q} \rangle_{\mathcal{H}_k} \leq \|f\|_{\mathcal{H}_k} \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}$, with equality if and only if $f = \alpha(\mu_\mathbb{P} - \mu_\mathbb{Q})$ for some $\alpha > 0$. The unique maximiser on the unit ball is $f^* = \frac{\mu_\mathbb{P} - \mu_\mathbb{Q}}{\|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}}$, giving $\sup_{\|f\|_{\mathcal{H}_k} \leq 1} \langle f, \mu_\mathbb{P} - \mu_\mathbb{Q} \rangle_{\mathcal{H}_k} = \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}$. Thus

$$\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}.$$

□

2.5.3 Proposition. If the kernel k of the RKHS $\mathcal{H}_k$ is characteristic, then $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = 0$ if and only if $\mathbb{P} = \mathbb{Q}$, and thus MMD defines a proper metric on the set of probability measures $\mathcal{P}$ on the measurable space $(\mathcal{X}, \mathcal{B}_\mathcal{X})$.

Proof. We know that $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k}$. By the properties of the norm $\|\cdot\|_{\mathcal{H}_k}$, $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = 0$ is equivalent to $\mu_\mathbb{P} = \mu_\mathbb{Q}$. Since the kernel k is characteristic, $\mu_\mathbb{P} = \mu_\mathbb{Q}$ if and only if $\mathbb{P} = \mathbb{Q}$. Thus $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = 0$ if and only if $\mathbb{P} = \mathbb{Q}$. □

The MMD thus measures the largest difference in expectations over functions in the unit ball of $\mathcal{H}_k$. The proposition above is used in this project to assess whether data generated from two random variables come from similar distributions.

As shown by Gretton et al. (2012), for X, X' drawn independently from $\mathbb{P}$ and Y, Y' drawn independently from $\mathbb{Q}$, we have

$$\mathrm{MMD}^2(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = \mathbb{E}_{X,X'}[k(X, X')] - 2\mathbb{E}_{X,Y}[k(X, Y)] + \mathbb{E}_{Y,Y'}[k(Y, Y')]$$

For estimation, let $\{x_i\}_{i=1}^n$ (resp. $\{y_i\}_{i=1}^m$) be independently and identically distributed samples from $\mathbb{P}$ (resp. $\mathbb{Q}$). The empirical kernel mean embeddings of $\mathbb{P}$ and $\mathbb{Q}$ are given by

$$\hat{\mu}_{\mathbb{P}} = \frac{1}{n}\sum_{i=1}^{n} k(\cdot, x_i), \qquad \hat{\mu}_{\mathbb{Q}} = \frac{1}{m}\sum_{j=1}^{m} k(\cdot, y_j) \tag{2.5.2}$$

These estimators yield the following estimate of the MMD

$$\begin{aligned}\widehat{\mathbf{MMD}}_b^2[\mathcal{H}, \mathbb{P}, \mathbb{Q}] &= \|\hat{\mu}_{\mathbb{P}} - \hat{\mu}_{\mathbb{Q}}\|_{\mathcal{H}}^2 \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} k(x_i, x_j) - \frac{2}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} k(x_i, y_j) + \frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m} k(y_i, y_j)\end{aligned} \tag{2.5.3}$$

This estimator is called the V-statistic and is biased; the bias arises from the diagonal terms $k(x_i, x_i)$ and $k(y_j, y_j)$. The unbiased variant, called the U-statistic, is defined by

$$\widehat{\mathbf{MMD}}_u^2(X, Y) = \frac{1}{m(m-1)}\sum_{i\neq j}^{m} k(x_i, x_j) - \frac{2}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} k(x_i, y_j) + \frac{1}{n(n-1)}\sum_{i\neq j}^{n} k(y_i, y_j) \tag{2.5.4}$$

In this project, we compare distributions of multivariate random variables across different domains to assess where statistical properties hold and where they fail. We use MMD for this comparison. The two MMD estimators in Eqs. (2.5.4) and (2.5.3) have time complexity $\mathcal{O}(dN^2)$ and storage complexity $\mathcal{O}(N^2)$, where d denotes the dimension of X and $N = \max(n, m)$. A more efficient estimator is therefore required, one that scales linearly in the number of samples.

For a continuous, positive definite, shift-invariant kernel k on $\mathbb{R}^d$, the kernel function can be approximated efficiently using Random Fourier Features (Rahimi and Recht, 2007). Rahimi and Recht exploit Bochner's theorem, which states that a continuous, positive definite, shift-invariant kernel k is positive definite if and only if the function $\tilde{k}$ defined by $\tilde{k}(x - y) = k(x, y)$ for all $x, y \in \mathbb{R}^d$ is the Fourier transform of a non-negative finite Borel measure. Assuming the kernel is properly scaled such that $\tilde{k}(\mathbf{0}) = 1$, this measure normalises to a proper probability density function $p(\boldsymbol{\omega})$, allowing us to express the kernel as an expectation

$$\begin{aligned}k(x, y) = \tilde{k}(x - y) &= \int_{\mathbb{R}^d} e^{i\boldsymbol{\omega}^T(x-y)} p(\boldsymbol{\omega})\, d\boldsymbol{\omega} \\ &= \mathbb{E}_{\boldsymbol{\omega}\sim p(\boldsymbol{\omega}), b\sim \mathrm{Unif}[0,2\pi]}\left[\sqrt{2}\cos\left(\boldsymbol{\omega}^T x + b\right) \cdot \sqrt{2}\cos\left(\boldsymbol{\omega}^T y + b\right)\right].\end{aligned} \tag{2.5.5}$$

The probability density function p is also the Fourier transform of $\tilde{k}$. For the Radial Basis Function (RBF) kernel define by $k(x, y) = \exp\left(-\frac{\|x-y\|_2^2}{2\sigma^2}\right)$ for all $x, y \in \mathbb{R}^d$, p is the probability density function of a zero-mean multivariate Gaussian distribution $\mathcal{N}(\mathbf{0}, \sigma^{-2}\boldsymbol{I}_d)$.

To construct the explicit Random Fourier Feature map $z : \mathbb{R}^d \to \mathbb{R}^D$, we draw D independent samples

$$(\boldsymbol{\omega}_j)_{j=1}^D \overset{\text{iid}}{\sim} p(\boldsymbol{\omega}), \qquad (b_j)_{j=1}^D \overset{\text{iid}}{\sim} \mathrm{Unif}[0,\, 2\pi], \tag{2.5.6}$$

and define each component of the randomised feature vector for $j = 1, 2, \ldots, D$ as

$$z(x)_j = \sqrt{\tfrac{2}{D}} \cos\Big(\boldsymbol{\omega}_j^\top x + b_j\Big) \quad , x \in \mathbb{R}^d. \tag{2.5.7}$$

The inner product $z(x)^\top z(y)$ is an unbiased estimator of $k(x, y)$ since $\mathbb{E}\left[z(x)^\top z(y)\right] = k(x, y)$. This approximation is then used to compute the empirical mean embeddings of the two distributions, and the MMD is obtained as the norm of their difference. The mean embeddings of $\mathbb{P}$ and $\mathbb{Q}$ are estimated by

$$\hat{\mu}_{\mathbb{P}} = \frac{1}{n} \sum_{i=1}^{n} z(x_i),$$
$$\hat{\mu}_{\mathbb{Q}} = \frac{1}{m} \sum_{j=1}^{m} z(y_j).$$

$$\widehat{\mathrm{MMD}}^2_{RFF}(\mathbb{P}, \mathbb{Q}, \mathcal{H}_k) = \|\hat{\mu}_{\mathbb{P}} - \hat{\mu}_{\mathbb{Q}}\|^2_{\mathcal{H}_k} \tag{2.5.8}$$

Thus, the MMD estimation with RFF has linear time complexity $\mathcal{O}(ND)$, where D is the embedding dimension to be set by the practitioner and $N = \max(n, m)$ is the maximum sample size of X and Y.

2.6 Conditional Mean Embeddings (CME)

We have defined the mean embedding of a distribution. We now extend this to conditional distributions. Let $(\mathcal{X}, \mathcal{B}_{\mathcal{X}})$ and $(\mathcal{Y}, \mathcal{B}_{\mathcal{Y}})$ be measurable spaces and let X and Y be two random variables taking values in $\mathcal{X}$ and $\mathcal{Y}$ respectively. Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ and $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ be positive definite kernels with corresponding RKHSs $\mathcal{H}_k$ and $\mathcal{G}_\ell$. Let $\mathbb{P}_X$ (resp. $\mathbb{P}_Y$) be the distribution of X (resp. Y); $\mathbb{P}_{Y|x}$ be the conditional distribution of Y given $X = x$. Let ϕ_k be the canonical feature map of the kernel k and ϕ_ℓ that of kernel ℓ.

2.6.1 Definition (Tensor Product Hilbert Space)**.** The tensor product space $\mathcal{H} = \mathcal{H}_k \otimes \mathcal{G}_\ell$ is defined as the metric completion of the algebraic tensor product spanned by all finite linear combinations of rank-1 elementary tensors $(f \otimes g)$, where $f \in \mathcal{H}_k$ and $g \in \mathcal{G}_\ell$. For any pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$, a rank-1 tensor $f \otimes g$ is evaluated as

$$(f \otimes g)(x, y) = f(x)g(y) \tag{2.6.1}$$

The inner product on $\mathcal{H}_k \otimes \mathcal{G}_\ell$ is uniquely induced by the marginal inner products via the relation

$$\langle f_1 \otimes g_1, f_2 \otimes g_2 \rangle_{\mathcal{H}_k \otimes \mathcal{G}_\ell} = \langle f_1, f_2 \rangle_{\mathcal{H}_k} \langle g_1, g_2 \rangle_{\mathcal{G}_\ell} \tag{2.6.2}$$

for all $f_1, f_2 \in \mathcal{H}_k$ and $g_1, g_2 \in \mathcal{G}_\ell$.

2.6.2 Definition (The Product Kernel)**.** The joint product kernel $h : (\mathcal{X} \times \mathcal{Y}) \times (\mathcal{X} \times \mathcal{Y}) \to \mathbb{R}$ associated with the tensor product space $\mathcal{H}_k \otimes \mathcal{G}_\ell$ is defined pointwise as:

$$h((x, y), (x', y')) = k(x, x')\, \ell(y, y') \quad \text{for all } (x, y) \in \mathcal{X} \times \mathcal{Y}, (x', y') \in \mathcal{X} \times \mathcal{Y}. \tag{2.6.3}$$

The joint product kernel h satisfies the reproducing property on $\mathcal{H}_k \otimes \mathcal{G}_\ell$.

2.6.3 Definition (Covariance Operator)**.** The covariance operator $C_{XX} : \mathcal{H}_k \to \mathcal{H}_k$ is the bounded linear operator defined by

$$C_{XX} := \mathbb{E}\left[\phi_k(X) \otimes \phi_k(X)\right] - \mu_X \otimes \mu_X. \tag{2.6.4}$$

2.6.4 Definition (Cross-Covariance Operator)**.** The cross-covariance operator $C_{YX} : \mathcal{H}_k \to \mathcal{G}_\ell$ is the Hilbert–Schmidt operator defined by

$$C_{YX} := \mathbb{E}\left[\phi_\ell(Y) \otimes \phi_k(X)\right] - \mu_Y \otimes \mu_X.$$

Following the logic for uncentred cross-covariance (Muandet et al., 2017, Section 3.2), we can show that $\langle g, C_{YX} f\rangle_{\mathcal{G}_\ell} = \mathrm{Cov}\left[g(Y)f(X)\right]$ for all f element of $\mathcal{H}_k$ and g element of $\mathcal{G}_\ell$. Indeed,

$$\begin{aligned}
\langle g, C_{YX} f\rangle_{\mathcal{G}_\ell} &= \Big\langle g, \mathbb{E}\left[\phi_\ell(Y) \otimes \phi_k(X)\right] f\Big\rangle_{\mathcal{G}_\ell} - \langle g, (\mu_Y \otimes \mu_X) f\rangle_{\mathcal{G}_\ell} \\
&= \mathbb{E}\left[\langle g, \phi_\ell(Y) \otimes \phi_k(X) f\rangle_{\mathcal{G}_\ell}\right] - \langle g, (\mu_Y \otimes \mu_X) f\rangle_{\mathcal{G}_\ell} \\
&= \mathbb{E}\left[\Big\langle g, \langle f, \phi_k(X)\rangle_{\mathcal{H}_k} \phi_\ell(Y)\Big\rangle_{\mathcal{G}_\ell}\right] - \Big\langle g, \langle f, \mu_X\rangle_{\mathcal{H}_k} \mu_Y\Big\rangle_{\mathcal{G}_\ell} \\
&= \mathbb{E}\left[\langle g, \phi_\ell(Y)\rangle_{\mathcal{G}_\ell} \langle f, \phi_k(X)\rangle_{\mathcal{H}_k}\right] - \langle g, \mu_Y\rangle_{\mathcal{G}_\ell} \langle f, \mu_X\rangle_{\mathcal{H}_k} \\
&= \mathbb{E}\left[g(Y)f(X)\right] - \mathbb{E}\left[g(Y)\right]\mathbb{E}\left[f(X)\right].
\end{aligned}$$

The last equality follows from the reproducing property of the RKHS and is the covariance of $f(X)$ and $g(Y)$.

The theorem below gives sample-based estimators for the cross-covariance and covariance operators. Let $\{(x_i, y_i)\}_{i=1}^n$ be sample of i.i.d. random variables $\{(X_i, Y_i)\}_{i=1}^n$ having distribution $\mathbb{P}_{XY}$. Let $\Phi = [\phi_k(x_1) \dots \phi_k(x_n)]^\top$ and $\Psi = [\phi_\ell(y_1) \dots \phi_\ell(y_n)]^\top$, be the feature matrices, $\mathbf{k}_x = (k(x, x_1) \cdots k(x, x_n))^\top$, $\mathbf{l}_y = (l(y, y_1) \cdots l(y, y_n))^\top$, and let $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ be the centring matrix, $\mathbf{K}_x = \Phi^\top\Phi$ and $\mathbf{L}_y = \Psi^\top\Psi$ be the Gram matrix. $[\mathbf{K}_x]_{ij} = k(x_i, x_j)$ and $[\mathbf{L}_y]_{ij} = \ell(y_i, y_j)$ for $i, j = 1, 2, \cdots, n$

2.6.5 Theorem (Empirical Covariance and Cross-Covariance (Song et al., 2009; Muandet et al., 2017))**.** *The empirical estimators of C_{XX} and C_{YX} are*

$$\widehat{C}_{XX} = \frac{1}{n}\,\Phi\mathbf{H}\Phi^\top = \frac{1}{n}\sum_{i=1}^n \Big(\phi_k(x_i) - \hat{\mu}_X\Big) \otimes \Big(\phi_k(x_i) - \hat{\mu}_X\Big),$$

$$\widehat{C}_{YX} = \frac{1}{n}\,\Psi\mathbf{H}\Phi^\top = \frac{1}{n}\sum_{i=1}^n \Big(\phi_\ell(y_i) - \hat{\mu}_Y\Big) \otimes \Big(\phi_k(x_i) - \hat{\mu}_X\Big),$$

with empirical mean embeddings $\hat{\mu}_X = \frac{1}{n}\sum_{i=1}^n \phi_k(x_i)$ and $\hat{\mu}_Y = \frac{1}{n}\sum_{i=1}^n \phi_\ell(y_i)$.

2.6.6 Definition (Conditional Mean Embedding (CME))**.** The conditional mean embedding of Y given $X = x$ is the element $\mu_{Y|x}$ of the RKHS $\mathcal{G}_\ell$ defined by

$$\mu_{Y|x} := \int_{\mathcal{Y}} \phi_\ell(y) \mathrm{d}\mathbb{P}_{Y|x}(y) \tag{2.6.5}$$

Conditional embeddings extend mean embeddings to conditional distributions (Klebanov et al., 2020).

We can define an operator such that the conditional mean embedding is obtained by applying the operator to the feature map.

2.6.7 Definition (Conditional Embedding Operator (Song et al., 2009))**.** The conditional embedding operator $\mathcal{U}_{Y|X} : \mathcal{H}_k \to \mathcal{G}_\ell$ is defined as

$$\mathcal{U}_{Y|X} := C_{YX} C_{XX}^{-1}, \tag{2.6.6}$$

2.6.8 Theorem ((Song et al., 2009, Thm. 4))**.** *Assume that for all $g \in \mathcal{G}_\ell$, the conditional expectation $\mathbb{E}[g(Y) \,|\, X]$ is an element of $\mathcal{H}_k$. Then the operator $\mathcal{U}_{Y|X}$ from Definition 2.6.7 satisfies both:*

(i) $\mathcal{U}_{Y|X}\phi_k(x) = \mathbb{E}[\phi_\ell(Y) \mid X = x] = \mu_{Y|x}$*, i.e. applying the operator to the feature map $\phi_k(x)$ yields the conditional mean embedding of $\mathbb{P}_{Y|x}$;*

(ii) $\langle g, \mu_{Y|x} \rangle_{\mathcal{G}_\ell} = \mathbb{E}[g(Y) \,|\, X = x]$ *for all $g \in \mathcal{G}_\ell$.*

The estimator of the conditional mean embedding is

$$\hat{\mu}_{Y|x} = \sum_{i=1}^{n} \beta_i \ell(y_i, \cdot) \tag{2.6.7}$$

where $\boldsymbol{\beta} = (\mathbf{K}_x + n\lambda \mathbf{I}_n)^{-1} \mathbf{k}_x$ (Muandet et al., 2017; Song et al., 2009).

3. Counterfactual Invariance

In observational studies with a causal structure, generating counterfactual predictions requires assumptions about the underlying causal model. If the SCM is known (or assumed), counterfactual predictions can be generated using the approach defined in Sec. 3.1 (Pearl, 2009). In the absence of a fully specified SCM, counterfactual generation often relies on matching or weighting schemes. Nearest-neighbour matching finds observations with the same values of Z but different values of A, enabling within-stratum comparisons. Inverse probability weighting (IPW) adjusts for selection bias when estimating causal effects. In this work, we only generate counterfactuals in a simulated setting. Moreover, generating a counterfactual requires recovering the latent noise variables that encode the individual's specific context in the abduction step in Pearl's three-step procedure. We require an injectivity assumption on the structural equations of the abduction variables with respect to the noise (Thm. 3.3.2), so that the noise can be uniquely recovered given the abduction variables.

3.1 Counterfactual Reasoning

As defined by Pearl (2009), the counterfactual distribution is obtained through three successive steps.

The first step is **Abduction**. Conditioning on observed evidence $W = w$, we obtain the posterior over exogenous variables $\mathbb{P}_{U|W=\mathbf{w}}$. The abduction induces a new SCM. The second step is **Intervention**. We modify the SCM via an intervention on $\mathbf{A}$, $\mathrm{do}(\mathbf{A} = \mathbf{a})$. The final step is **Prediction**. We propagate the intervened model to compute the counterfactual outcome. Formally,

3.1.1 Definition (Counterfactual Distribution)**.** Let $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, \mathbb{P}_U)$ be an SCM and let $\mathbf{A}, \mathbf{W} \subseteq \mathcal{V}$ be two node sets with observed values $W = \mathbf{w}$. Define the abducted SCM

$$\mathcal{M}^{\mathbf{w}} \;=\; (\mathcal{U},\, \mathcal{V},\, \mathcal{F},\, \mathbb{P}_{U|W=\mathbf{w}}) \tag{3.1.1}$$

The counterfactual distribution of an outcome $\mathbf{Y} \subseteq \mathcal{V}$ is defined as

$$\mathbb{P}_{Y^*_{\mathbf{a}}|W=\mathbf{w}} := \mathbb{P}_{Y|\mathrm{do}(\mathbf{A}=\mathbf{a}),\,\mathcal{M}^{\mathbf{w}}}$$

The left-hand side is read as "Interventional distribution of $\mathbf{Y}$ under $\mathrm{do}(\mathbf{A} = \mathbf{a})$ with respect to the abducted SCM $\mathcal{M}^{\mathbf{w}}$ ".

3.2 Independence Condition in RKHS

The task of measuring and enforcing independence constraints is fundamental to causal inference and fairness. Kernel methods provide flexible, non-parametric solutions that do not require distributional assumptions on the random variables (Pfister et al., 2018).

Let $(\mathcal{X}, \mathcal{B}_\mathcal{X})$ and $(\mathcal{Y}, \mathcal{B}_\mathcal{Y})$ be measurable spaces and let X and Y be two random variables taking values in $\mathcal{X}$ and $\mathcal{Y}$ respectively. Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ and $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ be positive definite kernels with corresponding RKHSs $\mathcal{H}_k$ and $\mathcal{G}_\ell$ such that they are separable. Let $\mathbb{P}_X$ (resp. $\mathbb{P}_Y$) be the distribution of X (resp. Y) and $\mathbb{P}_{XY}$ be the joint distribution.

3.2.1 Definition (Hilbert-Schmidt Independence Criterion (Gretton et al., 2005; Muandet et al., 2017))**.** The Hilbert–Schmidt Independence Criterion is a distribution-agnostic measure of statistical dependence

between X and Y, given by

$$\begin{aligned}\mathrm{HSIC}(X,Y;\,k,\ell) &:= \|C_{XY}\|_{HS}^2 \\ &= \mathbb{E}_{XX'YY'}\left[k(X,X')\ell(Y,Y')\right] + \mathbb{E}_{XX'}\left[k(X,X')\right]\mathbb{E}_{YY'}\left[\ell(Y,Y')\right] \\ &\quad - 2\mathbb{E}_{XY}\left[\mathbb{E}_{X'}\left[k(X,X')\right]\mathbb{E}_{Y'}\left[\ell(Y,Y')\right]\right],\end{aligned}$$

where (X,Y) and (X',Y') are independent and have the same distribution.

From a sample $\{(x_i,y_i)\}_{i=1}^n$ of i.i.d. random variables $\{(X_i,Y_i)\}_{i=1}^n$ having distribution $\mathbb{P}_{XY}$, we estimate HSIC by

$$\begin{aligned}\widehat{\mathrm{HSIC}}(X,Y,k,\ell) &= \|\hat{C}_{XY}\|_{\mathrm{HS}}^2 \\ &= \frac{1}{n^2}\sum_{i,j=1}^n k(x_i,x_j)\,\ell(y_i,y_j) + \frac{1}{n^4}\sum_{i,j=1}^n k(x_i,x_j)\sum_{p,q=1}^n \ell(y_p,y_q) \\ &\quad - \frac{2}{n^3}\sum_{i,j,p=1}^n k(x_i,x_j)\,\ell(y_i,y_p).\end{aligned} \tag{3.2.1}$$

Since the Hilbert–Schmidt norm $\|\cdot\|_{\mathrm{HS}}$ of a matrix is the Frobenius norm $\|\cdot\|_{\mathrm{F}}$,

$$\widehat{\mathrm{HSIC}}(X,Y;\,k,\ell) = \|\hat{C}_{XY}\|_{\mathrm{F}}^2 = \mathrm{tr}(\hat{C}_{XY}\hat{C}_{XY}^\top) \tag{3.2.2}$$

Moreover, with Thm. 2.6.5,

$$\begin{aligned}\widehat{\mathrm{HSIC}}(X,Y,k,\ell) &= \frac{1}{n^2}\,\mathrm{tr}(\Phi\mathbf{H}\Psi^\top\Psi\mathbf{H}\Phi^\top) \\ &= \frac{1}{n^2}\,\mathrm{tr}(\Psi^\top\Psi\mathbf{H}\Phi^\top\Phi\mathbf{H}) \\ &= \frac{1}{n^2}\,\mathrm{tr}(\mathbf{L}_y\mathbf{H}\mathbf{K}_x\mathbf{H})\end{aligned} \tag{3.2.3}$$

Gretton et al. (2005) define a similar estimator using $\frac{1}{(n-1)^2}$ as the prefactor of the trace. We use expression (3.2.3) to show that the function $f: Y \mapsto \widehat{\mathrm{HSIC}}(X,Y,k,\ell)$ is differentiable.

3.2.2 Proposition. Let $\frac{1}{n^2}\,\mathrm{tr}(\mathbf{L}_y\mathbf{H}\mathbf{K}_x\mathbf{H})$ be an estimator of $\mathrm{HSIC}(X,Y,k,\ell)$ as defined in Eq. (3.2.3). If the kernel ℓ is differentiable, then the estimation $\frac{1}{n^2}\,\mathrm{tr}(\mathbf{L}_y\mathbf{H}\mathbf{K}_x\mathbf{H})$ is differentiable with respect to y.

Proof. Let the kernel ℓ be differentiable. $\boldsymbol{\nabla}_y\mathbf{L}_y$ exists. Then

$$\begin{aligned}\boldsymbol{\nabla}_y\frac{1}{n^2}\,\mathrm{tr}(\mathbf{L}_y\mathbf{H}\mathbf{K}_x\mathbf{H}) &= \frac{1}{n^2}\,\boldsymbol{\nabla}_{\mathbf{L}_y}\,\mathrm{tr}(\mathbf{L}_y\mathbf{H}\mathbf{K}_x\mathbf{H})\,\boldsymbol{\nabla}_y\mathbf{L}_y \\ &= \frac{1}{n^2}\,\mathbf{H}^\top\mathbf{K}^\top\mathbf{H}^\top\boldsymbol{\nabla}_y\mathbf{L}_y \\ &= \frac{1}{n^2}\,\mathbf{H}\mathbf{K}_x\mathbf{H}\,\boldsymbol{\nabla}_y\mathbf{L}_y\end{aligned}$$

exists. □

3.2.3 Definition (Hilbert-Schmidt Conditional Independence Criterion (Park and Muandet, 2020; Quinzan et al., 2024))**.** Let X, Y, Z be random variables. The Hilbert–Schmidt Conditional Independence Criterion (HSCIC) between X and Y given Z is defined as the real-valued random variable

$$\mathrm{HSCIC}(X,\,Y\mid Z) = H_{X,\,Y|Z}\circ Z, \tag{3.2.4}$$

where $H_{X,Y|Z}$ is a real-valued function defined as

$$H_{X,Y|Z}(\mathbf{z}) := \|\mu_{\mathbb{P}_{XY|Z=\mathbf{z}}} - \mu_{\mathbb{P}_{X|Z=\mathbf{z}}} \otimes \mu_{\mathbb{P}_{Y|Z=\mathbf{z}}}\|_{\mathcal{H}_k \otimes \mathcal{G}_\ell}.$$

Let $\{(x_i, y_i, z_i)\}_{i=1}^n$ be a sample from $\{(X_i, Y_i, Z_i)\}_{i=1}^n \overset{\text{iid}}{\sim} \mathbb{P}_{XYZ}$. The estimate of $H^2_{X,Y|Z}(\mathbf{z})$ is given by

$$\begin{aligned}\hat{H}^2_{X,Y|Z}(\cdot) = \mathbf{h}_z^T(\cdot)\mathbf{W}_z(\mathbf{K}_x \odot \mathbf{K}_y)\mathbf{W}_z^T\mathbf{h}_z(\cdot) - 2\mathbf{h}_z^T(\cdot)\mathbf{W}_z((\mathbf{K}_x\mathbf{W}_z^T\mathbf{h}_z(\cdot)) \odot (\mathbf{K}_y\mathbf{W}_z^T\mathbf{h}_z(\cdot))) \\ + (\mathbf{h}_z^T(\cdot)\mathbf{W}_z\mathbf{K}_x\mathbf{W}_z^T\mathbf{h}_z(\cdot))(\mathbf{h}_z^T(\cdot)\mathbf{W}_z\mathbf{K}_y\mathbf{W}_z^T\mathbf{h}_z(\cdot))\end{aligned} \tag{3.2.5}$$

where $[\mathbf{K}_x]_{ij} = k(x_i, x_j)$, $[\mathbf{K}_y]_{ij} = \ell(y_i, y_j)$, $[\mathbf{K}_z]_{ij} = h(z_i, z_j)$ are Gram matrices and $\mathbf{h}_z(\cdot) = (h(z_1, \cdot) \cdots h(z_n, \cdot))^T$, $\mathbf{W}_z = (\mathbf{K}_z + n\lambda I_n)^{-1}$ and $\odot$ denotes element-wise multiplication of matrices (Park and Muandet, 2020). Following this definition, we have

3.2.4 Theorem (Conditional Independence (Park and Muandet, 2020)). *If the kernel of the RKHS $\mathcal{H}_k \otimes \mathcal{G}_\ell$ is characteristic, then $\mathrm{HSCIC}(X, Y \mid Z) = 0$ almost surely $\iff$ $X \perp\!\!\!\perp Y \mid Z$.*

3.3 Counterfactual Invariant Predictor

3.3.1 Definition (Counterfactual Invariance (Quinzan et al., 2024)). Let $\mathbf{A}$, $\mathbf{W}$ be (not necessarily disjoint) sets of nodes in a given SCM $\mathcal{M}$. Then $\mathbf{Y}$ is counterfactually invariant in $\mathbf{A}$ w.r.t. $\mathbf{W}$ if

$$\mathbb{P}_{Y^*_{\mathbf{a}}|W=\mathbf{w}}(y) = \mathbb{P}_{Y^*_{\mathbf{a}'}|W=\mathbf{w}}(y) \quad \text{almost surely,} \tag{3.3.1}$$

for all $\mathbf{a}, \mathbf{a}'$ belonging to the support of A and all $\mathbf{w}$ belonging to the support of W.

3.3.2 Theorem (Quinzan et al. (2024)). *Consider a causal graph $\mathcal{G}$ with node sets $\mathbf{A}, \mathbf{W}$ satisfying $(\mathbf{A} \cup \mathbf{W}) \cap \mathbf{Y} = \emptyset$, and let $\mathbf{Z}$ be a valid adjustment set for $(\mathbf{A} \cup \mathbf{W}, \mathbf{Y})$. For each $X \in \mathbf{W} \setminus \mathbf{A}$, let $g_X(\mathrm{Pa}(X), U_X)$ denote its structural equation with $\mathrm{Pa}(X) \subseteq \mathbf{A} \cup \mathbf{W}$, and assume g_X is injective in U_X. Then, across all SCMs compatible with $\mathcal{G}$, any predictor $\hat{Y}$ satisfying $\hat{Y} \perp\!\!\!\perp (A, W) \mid Z$ is counterfactually invariant in $\mathbf{A}$ with respect to $\mathbf{W}$.*

Theorem 3.3.2 gives the statistical conditions needed to verify whether a predictor $\hat{Y}$ is counterfactually invariant. Moreover, by definition of HSCIC (Def. 3.2.3), we have

3.3.3 Corollary. Whenever $\mathrm{HSCIC}(\hat{Y}, (A, W) \mid Z) = 0$ almost surely, the predictor $\hat{Y}$ achieves counterfactual invariance in A with respect to W, where (A, W) is the random variable associated with the node set $\mathbf{A} \cup \mathbf{W}$.

$\mathrm{HSCIC}(\hat{Y}, (A, W)|Z)$ is estimated from observational data using Eq. (3.2.5) with no assumption on the distribution or the Structural Causal Model. In a simulation setting, as we discuss in the next chapter, we can generate counterfactuals and assess the counterfactual invariance of the predictor. Quinzan et al. (2024) propose the Variance of Counterfactuals.

$$\mathrm{VCF}(\hat{Y}) := \mathbb{E}_{\mathbf{w}\sim\mathbb{P}_W}\left[\mathrm{Var}_{\mathbf{a}'\sim\mathbb{P}_A}\left[\mathbb{E}_{\hat{Y}^*_{\mathbf{a}'}|W=\mathbf{w}}[\hat{y}]\right]\right]. \tag{3.3.2}$$

A non-negative random variable has zero expected value if and only if it vanishes almost surely. So if VCF is zero, then for all $\mathbf{w}$ in the support of W, $\mathrm{Var}_{\mathbf{a}'\sim\mathbb{P}_A}\left[\mathbb{E}_{\hat{Y}^*_{\mathbf{a}'}|W=\mathbf{w}}[\hat{y}]\right] = 0$, which implies that the quantity $\mathbb{E}_{\hat{Y}^*_{\mathbf{a}'}|W=\mathbf{w}}[\hat{y}]$ is constant with respect to $\mathbf{a}' \sim \mathbb{P}_A$. In other words,

3.3.4 Proposition. The predictor $\hat{Y}$ achieves counterfactual invariance in $\mathbf{A}$ with respect to $\mathbf{W}$ if and only if its Variance of Counterfactuals $\mathrm{VCF}(\hat{Y}) = 0$ almost surely.

Let $\mathbf{X}$, $\mathbf{A}$, $\mathbf{Z}$, $\mathbf{W}$, $\mathbf{Y}$ be sets of nodes representing random variables. Let X be a random variable taking values in $\mathcal{X}$ and representing the input features, and Y be the target random variable taking values in $\mathcal{Y}$. Let $\mathcal{F}$ be a family of functions from $\mathcal{X}$ to $\mathcal{Y}$, and let $\hat{Y} = f(X)$ be a predictor of Y for $f \in \mathcal{F}$. We seek a function $f \in \mathcal{F}$ that is counterfactually invariant in $\mathbf{A}$ with respect to $\mathbf{W}$, i.e., $\mathbb{P}_{\hat{Y}^*_{\mathbf{a}}|W=\mathbf{w}}(y) = \mathbb{P}_{\hat{Y}^*_{\mathbf{a}'}|W=\mathbf{w}}(y)$ almost surely, for all $\mathbf{a}, \mathbf{a}'$ in the support of A and all $\mathbf{w}$ in the support of W. Under the assumptions of Thm. 3.3.2 and Cor. 3.3.3, the predictor must satisfy $\mathrm{HSCIC}(f(X), (A, W) \mid Z) = 0$ almost surely.

To find this function, Quinzan et al. (2024) propose adding HSCIC as a regulariser term whose strength is controlled by a trade-off parameter β. Let $L : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ be a task-specific loss function. The Counterfactual Invariance Predictor functional risk $\mathcal{R}$ is defined as

$$\mathcal{R}_{CIP}[f] = \mathbb{E}_{(X,Y)}\left[L(f(X), Y)\right] + \beta\, \mathrm{HSCIC}(f(X), (A, W) \mid Z). \tag{3.3.3}$$

The empirical training loss is therefore

$$\mathcal{L}_{CIP}(f(X), Y) = \mathcal{L}(f(X), Y) + \beta\, \widehat{\mathrm{HSCIC}}(f(X), (A, W) \mid Z), \tag{3.3.4}$$

where $\mathcal{L}(f(X), Y)$ is an empirical estimate of $\mathbb{E}_{(X,Y)}\left[L(f(X), Y)\right]$, and the HSCIC term is estimated using Eq. (3.2.5). To use gradient-based optimisation, the estimator $\widehat{\mathrm{HSCIC}}(f(X), (A, W) \mid Z)$ must be differentiable with respect to the model parameters. As with HSIC (Prop. 3.2.2), one can see from Eq. (3.2.5) that $\hat{H}^2_{X, f(X)|Z}(\cdot)$ is differentiable with respect to f whenever the kernel ℓ is differentiable. By the chain rule, it is then differentiable with respect to the model parameters. The Radial Basis Function kernel is differentiable, whereas the Laplacian kernel is not; we therefore use the RBF kernel throughout this project.

4. Methodology

Chapter 3 establishes that a predictor satisfying $\mathrm{HSCIC}(\hat{Y}, (A, W) \mid Z) = 0$ almost surely is counterfactually invariant in $\mathbf{A}$ with respect to $\mathbf{W}$ (Cor. 3.3.3), and that minimising HSCIC as a regularisation term is a tractable way to enforce this constraint during training. This chapter describes the experiments designed to evaluate this approach empirically.

The experiments are conducted in two settings. In the synthetic setting (Sec. 4.1), the SCM $\mathcal{M}$ is fully specified, so exogenous noise variables can be recovered and true counterfactuals can be generated. This allows us to compute the Variance of Counterfactuals $\widehat{\mathrm{VCF}}$ (Eq. (4.4.7)) and compare it directly with the estimated $\widehat{\mathrm{HSCIC}}$. Because HSCIC is estimated from observational data while VCF requires access to the SCM, they measure invariance through different routes; a key question is whether they track each other as the trade-off parameter β varies. In the real-data setting (Sec. 4.2), the SCM is unknown and counterfactual generation is not available, so HSCIC is the sole operational criterion. The ACS dataset is used to test whether the regularisation reduces HSCIC without causing a catastrophic drop in AUC-ROC, and whether the trade-off is stable across out-of-domain states.

Domain shift is common to both settings. The distribution of A at test time may differ from training, and a predictor that relies on spurious correlations between (A, W) and Y is expected to degrade as this shift grows. We use Maximum Mean Discrepancy (MMD) to rank out-of-domain domains by their distance from the training distribution, and examine how VCF, HSCIC, and AUC-ROC evolve across this ranking. The experiments specifically probe the regime of high MMD, where a non-invariant predictor is most exposed.

The following hypotheses structure the analysis, corresponding to the research questions of Sec. 1.2. First, HSCIC and VCF should decrease together as β increases in the synthetic setting, confirming that minimising the observational criterion also reduces the true counterfactual variance. Second, increasing β should reduce HSCIC at the cost of AUC, and a range of β values should exist where the invariance gain is meaningful without severe performance degradation. Third, retaining (A, W) in the input and penalising their use (Strategy 1, Strategy 2) should achieve lower HSCIC than deleting them entirely (Baseline 1), while matching or exceeding Baseline 1 on AUC-ROC, since deletion does not prevent the model from recovering the dependence through proxy features in L and discards the causal signal that (A, W) carry. Fourth, the warm-up strategy (Strategy 2) should achieve a more stable trade-off than joint training from epoch 0 (Strategy 1), because the task loss has converged before the invariance constraint is applied. Fifth, CIP-trained models should maintain lower VCF and HSCIC than the unregularised baseline in high-MMD domains, where the shift in A is largest.

4.1 Synthetic Data

We define an SCM $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, \mathbb{P}_U)$ that satisfies the assumptions of Thm. 3.3.2. We denote by $f_V \in \mathcal{F}$ the structural equation of V and by U_V its noise. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be the associated Causal DAG.

4.1.1 Node Roles

Let $Z = (Z_1, Z_2, Z_3)$ denote the adjustment set, $A = (A_0, A_1, A_2)$ the counterfactual variable, $W = (W_1, W_2, W_3)$ the abduction variable, and L_1, L_2 other random variables. Thm. 3.3.2 requires

$$\mathrm{Pa}(W_i) \subseteq \mathbf{A} \cup \mathbf{W}, \quad \text{for all } W_i \in \mathbf{W},$$

and the structural equation of W_i to be injective with respect to its exogenous variable for all $W_i \in \mathbf{W} \setminus \mathbf{A}$.

4.1.2 Structural Equations

Let $\tilde{A}$ be the random variable defined by $\tilde{A} := A_0 + A_1 + A_2$. The structural equations of the adjustment set $\mathbf{Z}$ are

$$Z_1 = U_{Z_1}, \tag{4.1.1}$$
$$Z_2 = \tanh(0.5\, Z_1 + U_{Z_2})\,, \tag{4.1.2}$$
$$Z_3 = U_{Z_3}. \tag{4.1.3}$$

where $U_{Z_1} \sim \mathrm{Bern}(0.5)$. The counterfactual variable components follow

$$A_0 = 1.5\, Z_1 + 1.2\, U_{A,0}, \tag{4.1.4}$$
$$A_1 = 2.0\, Z_2 + 1.5\, U_{A,1}, \tag{4.1.5}$$
$$A_2 = 1.0\, Z_3 + 1.8\, U_{A,2}. \tag{4.1.6}$$

The stable attributes are

$$W_1 = 0.6\, \tilde{A} + U_{W_1}, \tag{4.1.7}$$
$$W_2 = \sigma\Big(0.6\, \tilde{A} + U_{W_2}\Big)\,, \tag{4.1.8}$$
$$W_3 = \sigma\Big(0.5\, \tilde{A} + U_{W_3}\Big)\,. \tag{4.1.9}$$

The feature nodes are

$$L_1 = 0.8\, W_1 + 0.5\, W_2 + U_{L_1}, \tag{4.1.10}$$
$$L_2 = \tanh(0.7\, W_2 + 0.6\, W_3) + U_{L_2}. \tag{4.1.11}$$

Finally, the outcome is $Y = \sigma(\eta)$ with

$$\begin{aligned}\eta = {} & 0.1\, L_1 + 0.2\, L_2 + 0.2\, W_1 + 0.1\, W_2 + 0.1\, W_3 \\ & + 2.0\, Z_1 + 1.0\, Z_2 + 0.8\, Z_3 + 0.3\, \sigma(\tilde{A}) - 1.5 + U_Y,\end{aligned} \tag{4.1.12}$$

where $U_Y \sim \mathcal{N}(0,\, 2.2)$. The three structural equations of W_i are each injective in their respective noise: f_{W_1} is affine in U_{W_1}, and f_{W_2}, f_{W_3} compose the strictly monotone sigmoid σ with an affine shift in U_{W_2} and U_{W_3} respectively. The parent condition $\mathrm{Pa}(W_i) \subseteq \mathbf{A} \cup \mathbf{W}$ holds by construction, since each W_i depends only on $\tilde{A} = A_0 + A_1 + A_2$. The graph induced by the SCM $\mathcal{M}$ is shown in Fig. 4.1. $\mathbf{Z}$ satisfies Def. 2.2.5 as a valid adjustment set for $(\mathbf{A} \cup \mathbf{W}, Y)$, and all parents of nodes in $\mathbf{W}$ belong to $\mathbf{A} \cup \mathbf{W}$ by construction. This SCM therefore satisfies the assumptions of Thm. 3.3.2, so we can use HSCIC to verify whether a predictor of Y is counterfactually invariant.

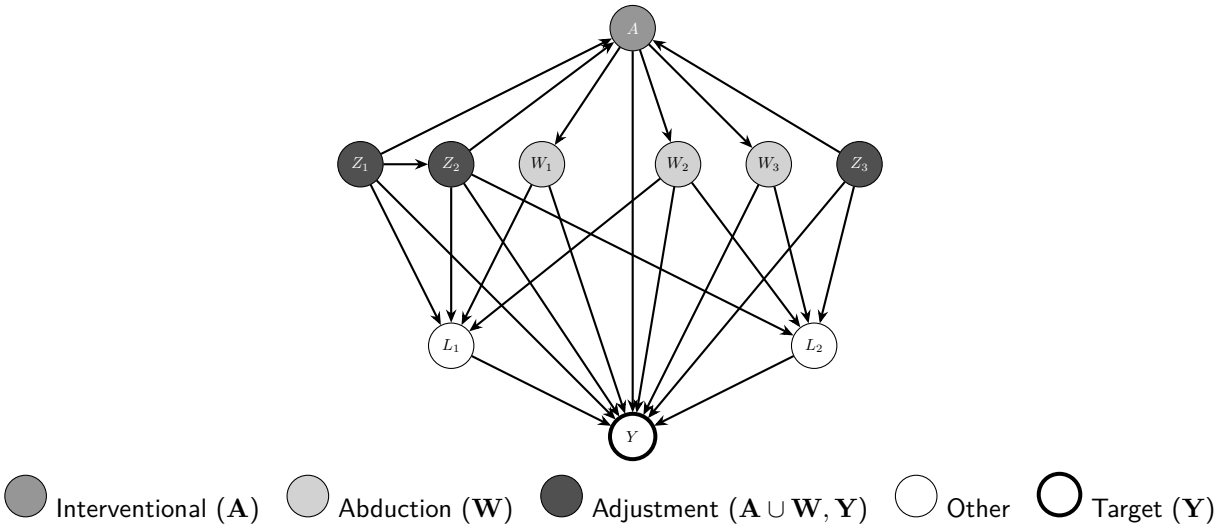


Figure 4.1: Causal DAG Induced by the simulated SCM $\mathcal{M}$.

4.1.3 Domain Shift

The noise of A is reparametrised per domain e as

$$U_A^{(e)} = b^{(e)}\, U_A^{(0)} + a^{(e)}, \tag{4.1.13}$$

with $(a^{(0)}, b^{(0)}) = (0, 1)$ for the reference domain. The confounder noise is not counter-shifted ($c^{(e)} = 0$). Domains 1–5 shift the mean of A (loc-type OOD); domains 6–9 expand its variance (scale-type OOD). W, L_1, L_2 are unshifted across all domains. The full domain shift simulation is given in Table 4.1.

Table 4.1: Domain-shift simulation.

Domain	Type	$a^{(e)}$	$b^{(e)}$
0	source	0.0	1.0
1	loc	+1.0	1.0
2	loc	+2.0	1.0
3	loc	+3.0	1.0
4	loc	+4.0	1.0
5	loc	+5.0	1.0
6	scale	0.0	1.5
7	scale	0.0	2.0
8	scale	0.0	2.5
9	scale	0.0	3.0

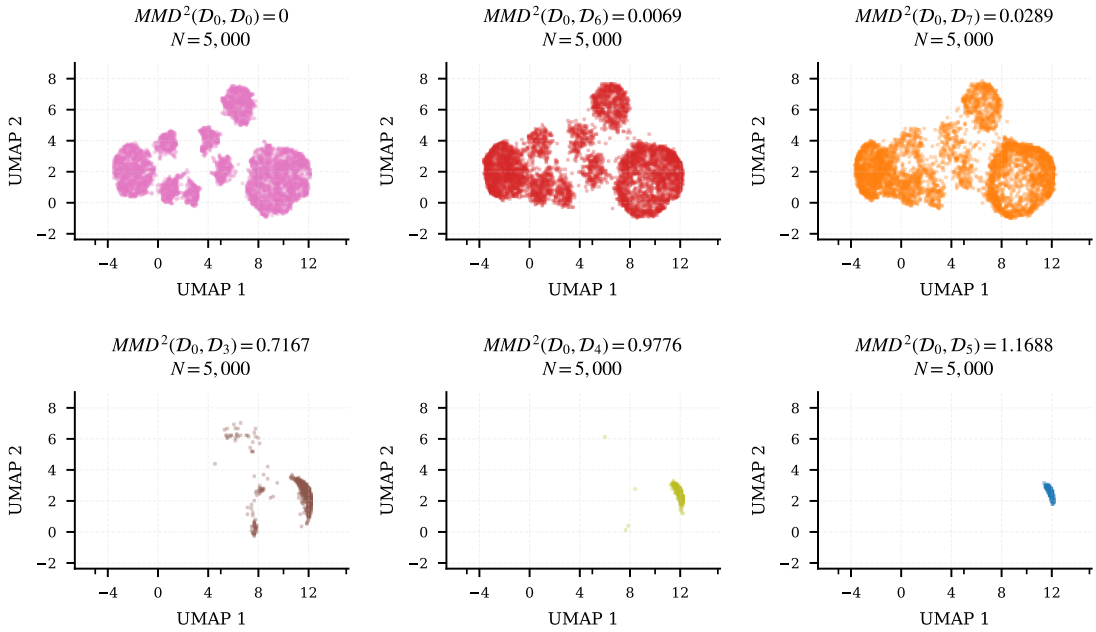


Figure 4.2: Synthetic data: UMAP projection of distribution (A, W, Y), (the train domain data, the 2 with lowest MMD and the 3 with highest MMD) ordered by MMD^2

Fig. 4.2 shows the UMAP projection of the distribution (A, W, Y) ordered by MMD^2.

4.2 Classification Task on Real Dataset

4.2.1 Dataset

Administered annually by the US Census Bureau, the American Community Survey (ACS) is a large-scale household survey accessible programmatically via the `folktables` Python library, which provides standardised task definitions and consistent feature-extraction pipelines across US states and survey years (Ding et al., 2021). We use data from 2018; the unit of observation is an individual aged above 16.

4.2.2 Prediction Tasks

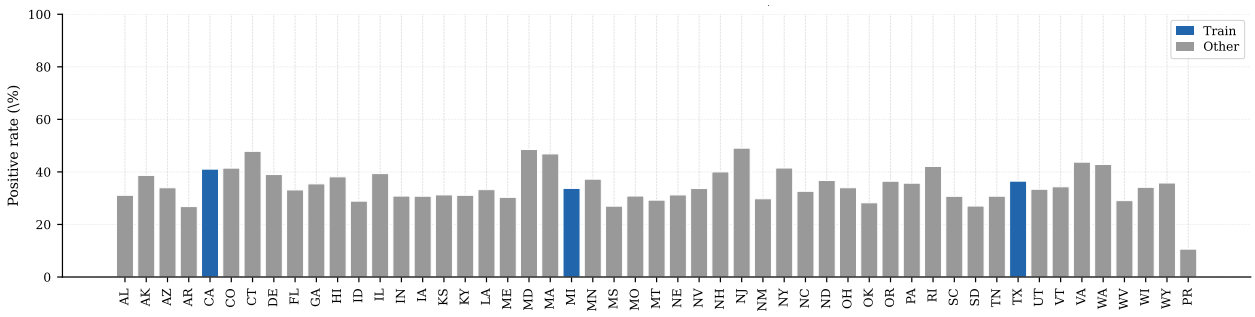


Figure 4.3: Positive rate per state for the Income task.

We consider two binary classification tasks. The first predicts if a respondent's annual income exceeds \$50,000 ($\mathrm{PINCP} > \$50,000$). The second classifies whether an individual has public health coverage (PUBCOV). As depicted in Figs. 4.3 and 4.4, the label distribution is broadly balanced across states.

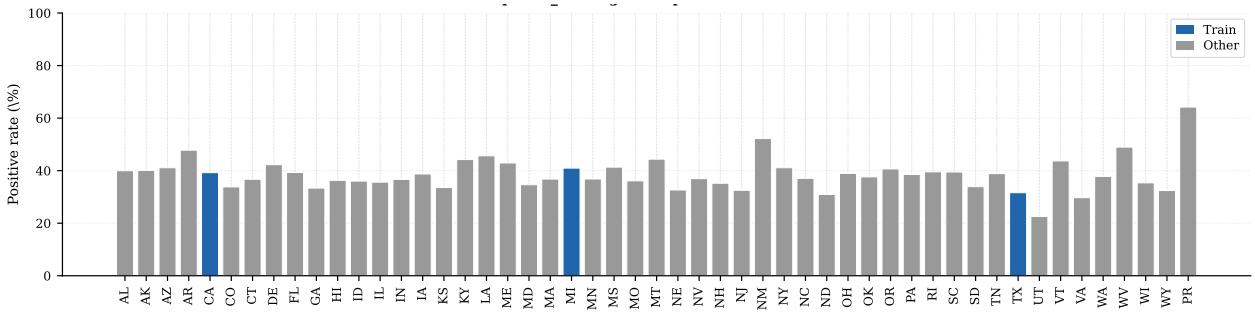


Figure 4.4: Positive rate per state for the Public Coverage task.

4.2.3 Domain Split

Each experiment trains on one US state (*train domain*) and evaluates on all remaining states (*OOD domains*). Three training domains are considered: Michigan (MI), California (CA), and Texas (TX). Fig. 4.5 shows the sample counts per state; those highlighted in blue serve as training domains. This cross-state evaluation protocol follows prior work on domain generalisation with the same survey data used by Nastl and Hardt (2024).They categorise the features into causal, arguably causal, and anti-causal. As with the synthetic data, we use MMD, estimated using Random Fourier Features, to rank the out-of-domain states by their distributional similarity to the training state.

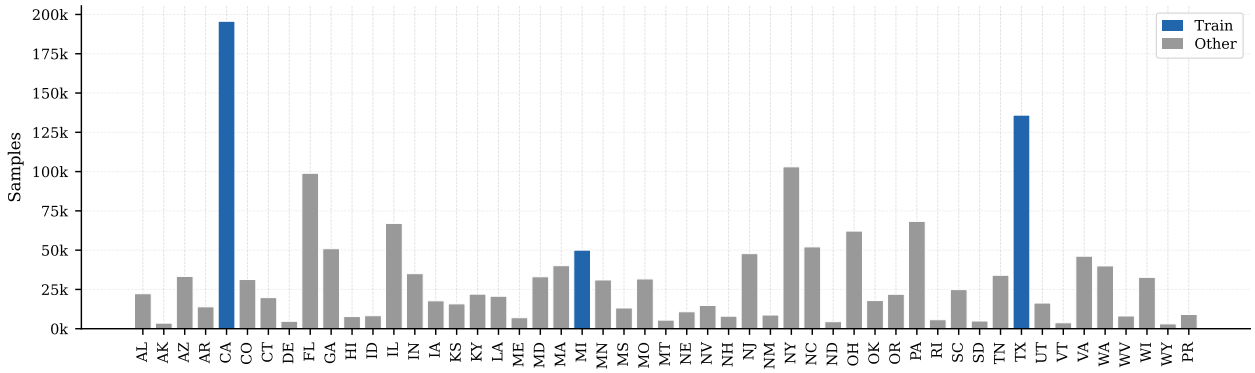


Figure 4.5: Sample counts per state. States highlighted in blue are used as training domains.

4.2.4 Causal Graph and Feature Partitions

For each task we specify a DAG encoding the assumed causal structure over the ACS features. The DAG determines the partition into counterfactual variables A, mediator/abduction variables W, adjustment set Z, remaining covariates L, and target Y. Three CIP types are considered for each classification task: **Arguably Causal**, **State Dependent**, and **Fairness**. These three types define the variables on which we would like the model to be invariant.

Income Task. Fig. 4.6 shows the three DAG variants for the Income task, and the corresponding feature assignments follow.

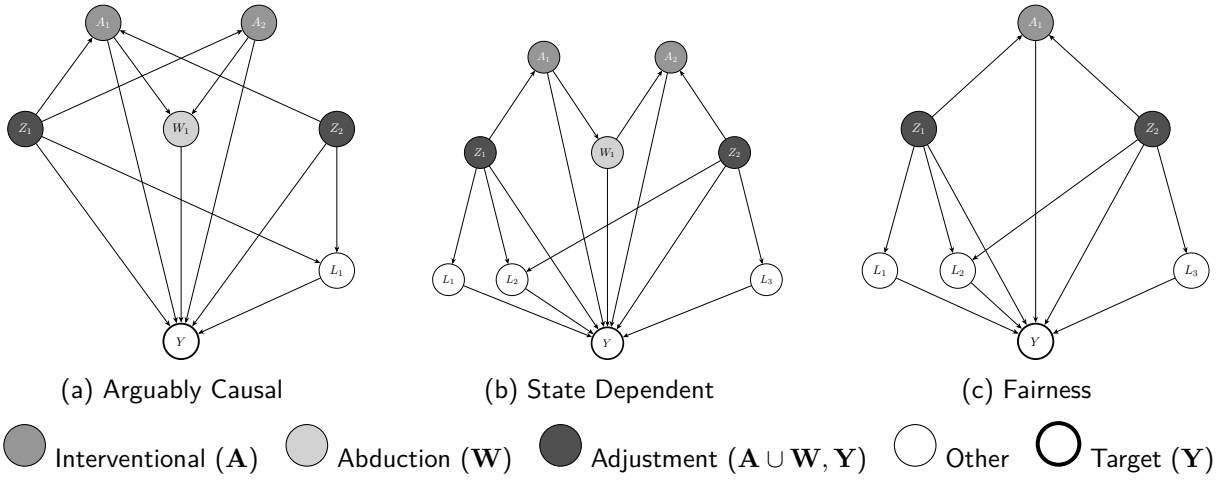


Figure 4.6: DAGs for the three CIP types (Income task).

Income: Arguably Causal. A_1: SCHL, CIT. A_2: ENG, NWLA, FER. W_1: OCCP, COW, WKHP, WKW, WRK. Z_1: AGEP, SEX, RAC1P, POBP. Z_2: ST, MAR. L_1: HINS1, HINS3, RELP.

Income: State Dependent. A_1: ST. A_2: WKHP, WKW. W_1: DIVISION, COW, OCCP. Z_1: AGEP, SEX, RAC1P, POBP. Z_2: SCHL. L_1: MAR, CIT. L_2: WRK, ENG, NWLA, FER. L_3: HINS1, HINS3, RELP.

Income: Fairness. A_1: SEX, RAC1P, POBP. Z_1: AGEP, ST. Z_2: MAR, CIT. L_1: SCHL, ENG, NWLA, FER. L_2: OCCP, COW, WKHP, WKW, WRK. L_3: HINS1, HINS3, RELP.

Public Coverage Task. Fig. 4.7 shows the three DAG variants for the Public Coverage task, and the corresponding feature assignments follow.

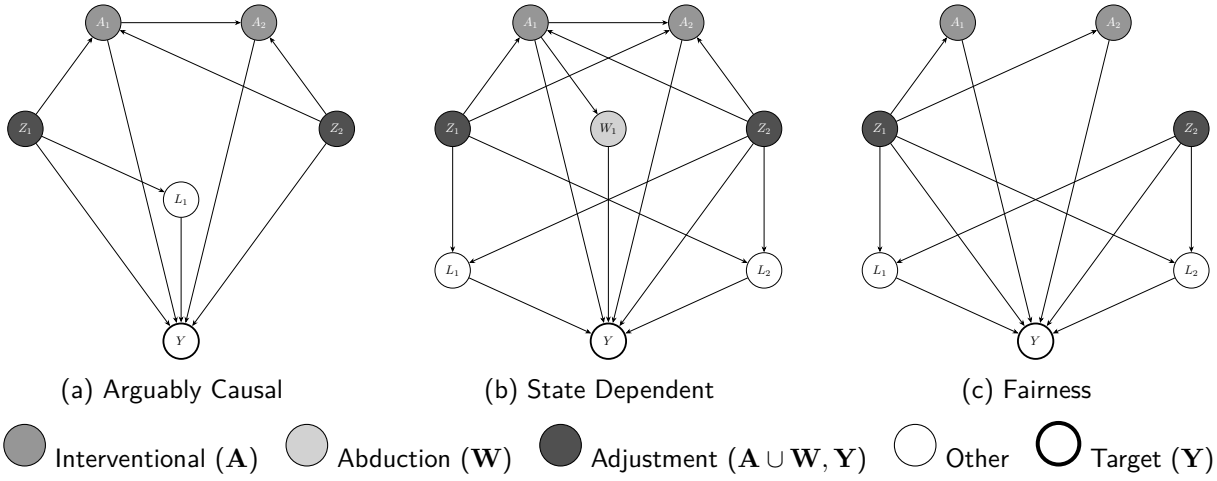


Figure 4.7: DAGs for the three CIP types (Public Coverage task).

Public Coverage: Arguably Causal. A_1: CIT, SCHL, MAR, FER, MIG. A_2: ESP, ESR, PINCP. Z_1: AGEP, SEX, RAC1P, ANC, NATIVITY. Z_2: DEAR, DEYE, DREM, ST. L_1: DIS.

Public Coverage: State Dependent. A_1: ST, MAR. A_2: ESR, PINCP. W_1: DIVISION. Z_1: AGEP,

SEX, RAC1P. Z_2: DIS, CIT, NATIVITY. L_1: DEAR, DEYE, DREM, SCHL. L_2: ESP, FER, MIG.

Public Coverage: Fairness. A_1: SEX, RAC1P. A_2: DEAR, DEYE, DREM. Z_1: AGEP, NATIVITY, ANC. Z_2: ST. L_1: DIS, MIG. L_2: CIT, SCHL, ESR, PINCP, MAR, ESP, FER.

4.3 Experiment Design

The end-to-end pipeline proceeds in five phases.

Phase 1: Domain ordering via MMD. We estimate the MMD between each candidate OOD state and the training state using Random Fourier Features. This produces a ranked ordering of OOD states by proximity to the training distribution. UMAP projections are applied to validate that the MMD-based ordering is coherent with the low-dimensional geometry of the data.

Phase 2: Data preprocessing. Features are partitioned into L, (A, W), Z, and Y. The full input feature vector is $X = (L, A, W, Z)$. All preprocessing is fitted on training-domain data and applied to out-of-domain data to prevent leakage.

Phase 3: Experiment execution. Four configurations are run per (task, CIP type, train domain, β) as shown in Table 4.2.

Phase 4: K-fold cross-state evaluation. Each state takes a turn as the held-out OOD domain; metrics are aggregated across folds.

Phase 5: Generating Counterfactuals. For the synthetic data setting, we use the trained model and the Structural Causal Model to generate counterfactuals and compute the VCF.

Table 4.2: Experiment configurations evaluated per (task, CIP type, train domain, β).

Name	Description
$\textsc{Baseline}\ 0$	Full X, no regularisation
$\textsc{Baseline}\ 1$	Remove $\{A, W\}$ from input features, no regularisation
$\textsc{Strategy}\ 1$	Full X, HSCIC penalty from epoch 0
$\textsc{Strategy}\ 2$	Full X, cross-entropy warm-up then HSCIC penalty

In $\textsc{Strategy}\ 2$, training proceeds in two stages. Stage 1 (epochs 1 to t_{warmup}): pure cross-entropy minimisation with early stopping on validation loss. Stage 2 (from epoch t_{warmup}): the term $+\beta \cdot \widehat{\mathrm{HSCIC}}$ is added to the cross-entropy.

4.3.1 Experiment Grid for ACS dataset

Table 4.3: Experiment grid for the ACS dataset: axes and values swept.

Axis	Values
Tasks	Income, Public Coverage
CIP feature type	*arguably_causal*, *fairness*, *state_dependent*
Train domains	CA, TX, MI
Experiments	Baseline 0, Baseline 1, Strategy 1, Strategy 2
β	1.0, 10.0, 100.0, 1000.0 (Strategy 1/Strategy 2 only)

4.4 Experiment Losses

Throughout this section, $f_\theta : \mathcal{X} \to (0,1)$ denotes a parametric function with parameters θ, $y \in \{0,1\}^n$ the vector of true labels, $\hat{y} = f_\theta(x) \in (0,1)^n$ the corresponding predicted probabilities, $\beta > 0$ a regularisation coefficient, and t the current training epoch. $\mathcal{L}_{\mathrm{CE}}(\hat{y}, y)$ denotes the binary cross-entropy loss, $\widehat{\mathrm{HSCIC}}(\hat{y}, (A,W) \mid Z)$ the empirical Hilbert–Schmidt Conditional Independence Criterion (HSCIC) penalty, and $\mathcal{L}_{\mathrm{B0}}$, $\mathcal{L}_{\mathrm{B1}}$, $\mathcal{L}_{\mathrm{S1}}$, $\mathcal{L}_{\mathrm{S2}}$ the per-experiment objective functions defined below.

4.4.1 Baseline 0: Full Features, No Regularisation

$$\mathcal{L}_{\mathrm{B0}}(\hat{y}, y) = \mathcal{L}_{\mathrm{CE}}(\hat{y}, y) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\Big]. \tag{4.4.1}$$

The model has access to all features X, including (A, W) and Z. No conditional independence constraint is imposed.

4.4.2 Baseline 1: Features Without (A, W), No Regularisation

$$\mathcal{L}_{\mathrm{B1}}(\hat{y}, y) = \mathcal{L}_{\mathrm{CE}}(\hat{y},\, y), \qquad \hat{y} = f_\theta(\tilde{X}), \tag{4.4.2}$$

where $\tilde{X}$ denotes the reduced input with the counterfactual and abduction variables (A, W) removed.

4.4.3 Strategy 1: CIP from Epoch 0

$$\mathcal{L}_{\mathrm{S1}}(\hat{y}, y) = \mathcal{L}_{\mathrm{CE}}(\hat{y}, y) + \beta \cdot \widehat{\mathrm{HSCIC}}(\hat{y}, (A,W) \mid Z). \tag{4.4.3}$$

HSCIC regularisation is applied from the first training step.

4.4.4 Strategy 2: Warm-up then CIP

$$\mathcal{L}_{\mathrm{S2}}(\hat{y}, y) = \begin{cases} \mathcal{L}_{\mathrm{CE}}(\hat{y}, y) & t \le t_{\mathrm{warmup}}, \\ \mathcal{L}_{\mathrm{CE}}(\hat{y}, y) + \beta \cdot \widehat{\mathrm{HSCIC}}(\hat{y}, (A,W) \mid Z) & t > t_{\mathrm{warmup}}, \end{cases} \tag{4.4.4}$$

where t is the current epoch and $t_{\text{warmup}} = 40$. We use the Radial Basis Function kernel, defined by

$$k(u, v) = \exp\left(-\frac{\|u - v\|^2}{2\sigma^2}\right), \tag{4.4.5}$$

with bandwidth σ set via the median heuristic (Garreau et al., 2018)

$$\sigma = \sqrt{\frac{1}{2}\,\text{median}\Big(\{\|x_i - x_j\|^2\}_{i \neq j}\Big)}. \tag{4.4.6}$$

4.4.5 Model Architecture

All experiments use a Multi-Layer Perceptron (MLP). Table 4.4 summarises the fixed architectural choices.

Table 4.4: MLP architecture.

Parameter	Value
Hidden layers	$[128,\ 64]$
Activation	GELU
Dropout	0.2
Layer normalisation	after each hidden layer
Output	1 logit $\rightarrow$ sigmoid
Task	binary classification

4.4.6 Optimiser

Training uses AdamW (Loshchilov and Hutter, 2019) with the hyperparameters shown in Table 4.5. AdamW decouples weight decay from the gradient update, unlike Adam, which folds ℓ_2 regularisation into the gradient and which would therefore conflate the ℓ_2 penalty with the HSCIC gradient signal. Since we do not penalise model weights directly but instead push the model to minimise the estimated independence criterion, AdamW provides consistent regularisation regardless of gradient magnitude.

Table 4.5: Optimiser hyperparameters.

Parameter	Value
Type	AdamW
Learning rate	10^{-4}
Weight decay	10^{-4}
(β_1, β_2)	$(0.9,\ 0.999)$

4.4.7 Training Loop

Table 4.6 lists the training loop parameters.

Table 4.6: Training loop parameters.

Parameter	Value
Max epochs	60
Batch size	1024
Gradient clipping	1.0 (global norm)
Mixed precision	disabled
Early stopping	patience $= 10$, mode $= \min$, $\delta = 10^{-4}$

4.4.8 Cross-Validation

A stratified K-fold scheme is applied to the in-distribution training domain (Table 4.7).

Table 4.7: Cross-validation setup.

Parameter	Value
K	5 folds
global seed	0
split seed	42
sampling seed	123
Test hold-out	15% (stratified, fixed before folding)
Validation	one fold of remaining 85%; rest $=$ train
Stratification	on binary target Y

Metrics are reported as mean $\pm$ std across folds.

4.4.9 Randomness Control

Three sources of randomness are controlled with separate seeds: data splitting, dataset generation or loading order, and model initialisation.

4.4.10 Metrics

All metrics are computed on the in-distribution train, validation, and test splits, as well as each OOD domain separately. The metrics reported are as follows.

- **AUC-ROC**: area under the receiver operating characteristic curve; primary performance metric for assessing discriminative power.
- **Macro-F1**: macro-averaged F1 score, which balances precision and recall across classes.
- $\widehat{\mathrm{HSCIC}}(\hat{Y}, (A, W) \mid Z)$: estimated HSCIC, our certificate of counterfactual invariance.

4.4.11 Synthetic Experiments: Additional Evaluation

For the synthetic experiment we compute the Variance of Counterfactual Predictions as defined in Eq. (3.3.2). Let u_i be the realisation of the noise U for the i-th individual, recovered from the data

using the known SCM. For k quantile-spaced intervention values $a_1, \ldots, a_k$ the estimator is

$$\widehat{\mathrm{VCF}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{k}\sum_{j=1}^{k}\left(\hat{p}_{i,j} - \bar{p}_i\right)^2, \qquad \mathrm{SE} = \frac{\hat{\sigma}_{\mathrm{VCF}}}{\sqrt{n}}, \tag{4.4.7}$$

where $\hat{p}_{i,j}$ is the model's predicted probability under intervention $\mathrm{do}(A = a_j)$ with all other noise fixed at u_i, and $\bar{p}_i = \frac{1}{k}\sum_{j=1}^{k}\hat{p}_{i,j}$. Computation parameters are given in Table 4.8.

Table 4.8: VCF computation parameters.

Parameter	Value	Description
n	1 000	individuals drawn per domain (outer expectation)
k	100	quantile-spaced A intervention values (inner variance)

The computation follows the three steps of counterfactual reasoning described in Sec. 3.1. VCF is computed post-training for each fold using the saved model checkpoint, feature preprocessor, and SCM, and then averaged across folds.

4.4.12 SHAP Analysis

For the model trained on the ACS datasets, SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017) is applied post-hoc using DeepExplainer with a background reference set drawn from in-distribution training data (Table 4.9).

Table 4.9: SHAP computation parameters.

Parameter	Value	Description
Background samples	100	reference set to marginalise out features
Explained samples (in-domain)	1 000	from test split of training domain
Explained samples (OOD)	100 per state	from each held-out state

SHAP values are computed for every experiment configuration and every β value, separately for the in-distribution test set and each out-of-domain dataset.

5. Results and Discussion

5.1 Synthetic Data

For the synthetic data, the SCM is known and constructed to satisfy the assumptions of Thm. 3.3.2, so we can recover the exogenous noise of each individual and generate true counterfactuals. Using the trained model, we compute the Variance of Counterfactuals (VCF) of Eq. (3.3.2) and compare it with the estimated HSCIC and AUC-ROC.

5.1.1 Counterfactual Invariance and Performance Trade-off

Figs. 5.1 and 5.2 plot VCF against the estimated HSCIC as β increases, for STRATEGY 1 and STRATEGY 2 respectively. Across all domain shifts, HSCIC decreases with β. The accompanying VCF tracks HSCIC closely in the low-to-moderate MMD^2 range, showing that minimising the HSCIC constraint also reduces the true counterfactual variance. Beyond $\beta = 100$, for STRATEGY 2, the VCF trend diverges from HSCIC; VCF rebounds while HSCIC continues falling, signalling that the finite-sample HSCIC approximation breaks down under extreme regularisation.

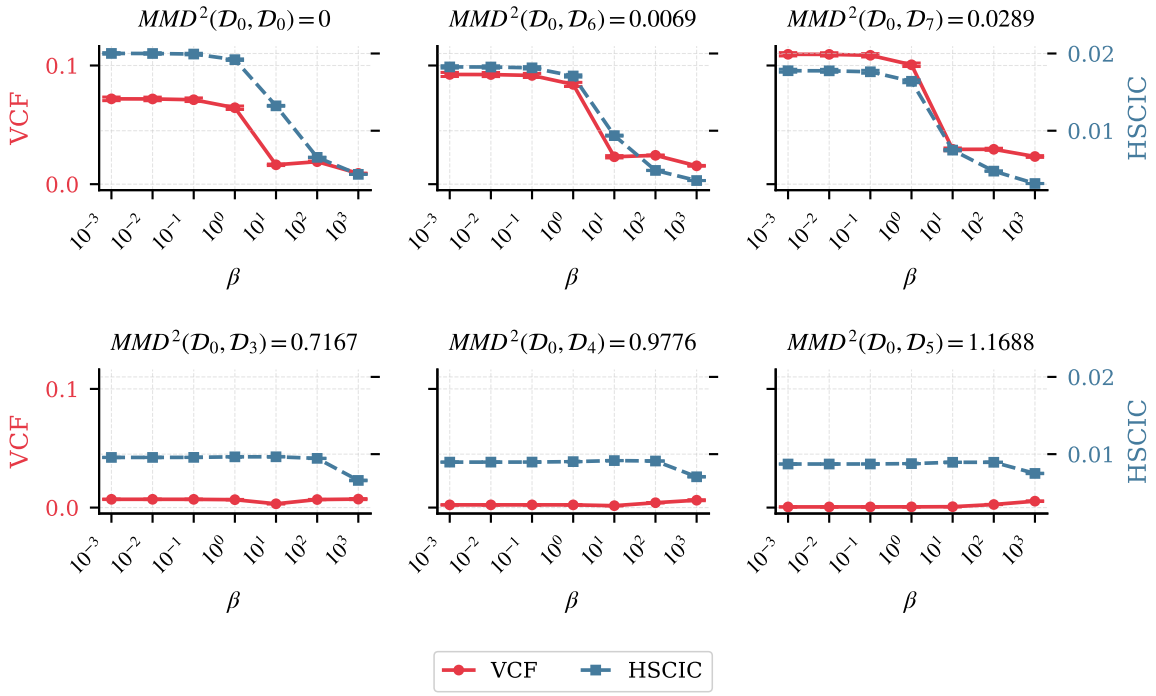


Figure 5.1: Synthetic: VCF and HSCIC versus β for STRATEGY 1 (joint optimisation) across the in-domain reference and shifted domains ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

The relationship between VCF and predictive performance is shown in Figs. 5.3 and 5.4. AUC-ROC is stable at low-to-moderate β and then degrades sharply. The trade-off parameter should be chosen

so that the trained model achieves lower HSCIC (and VCF) at minimal AUC drop; a threshold on the accepted performance drop can be used as a selection criterion (Quinzan et al., 2024).

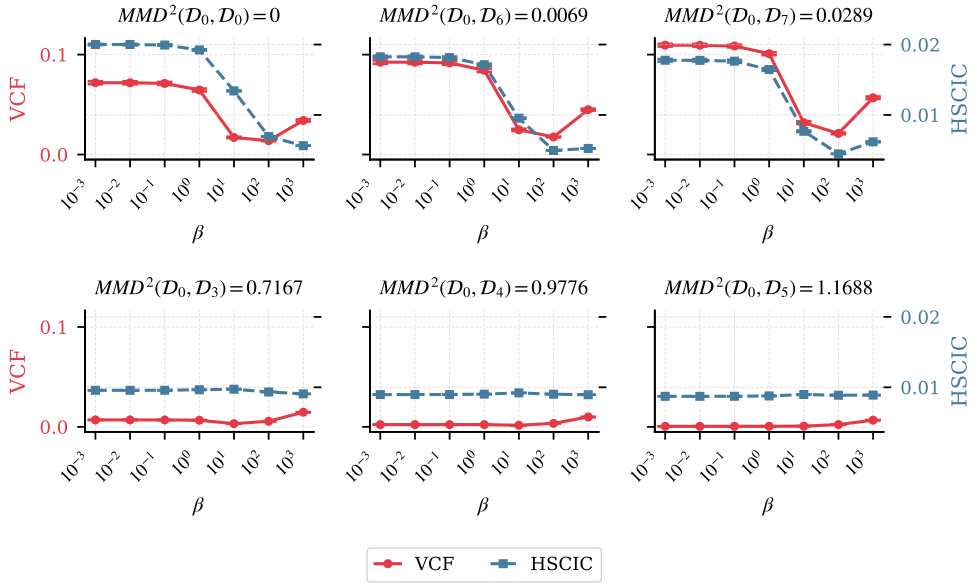


Figure 5.2: Synthetic: VCF and HSCIC versus β for STRATEGY 2 (warm-up then CIP) across the in-domain reference and shifted domains ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

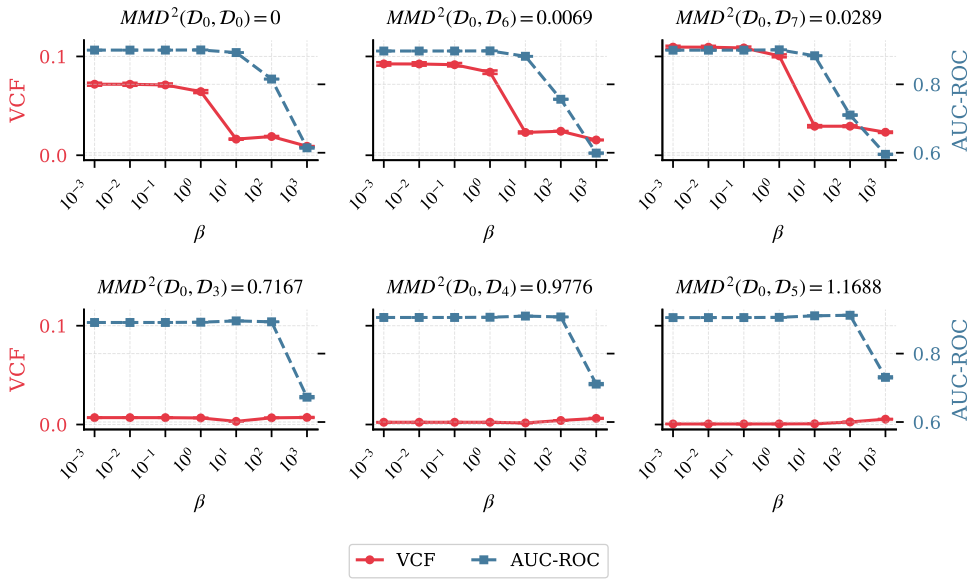


Figure 5.3: Synthetic: VCF and AUC-ROC versus β for STRATEGY 1 (joint optimisation) across the in-domain reference and shifted domains ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

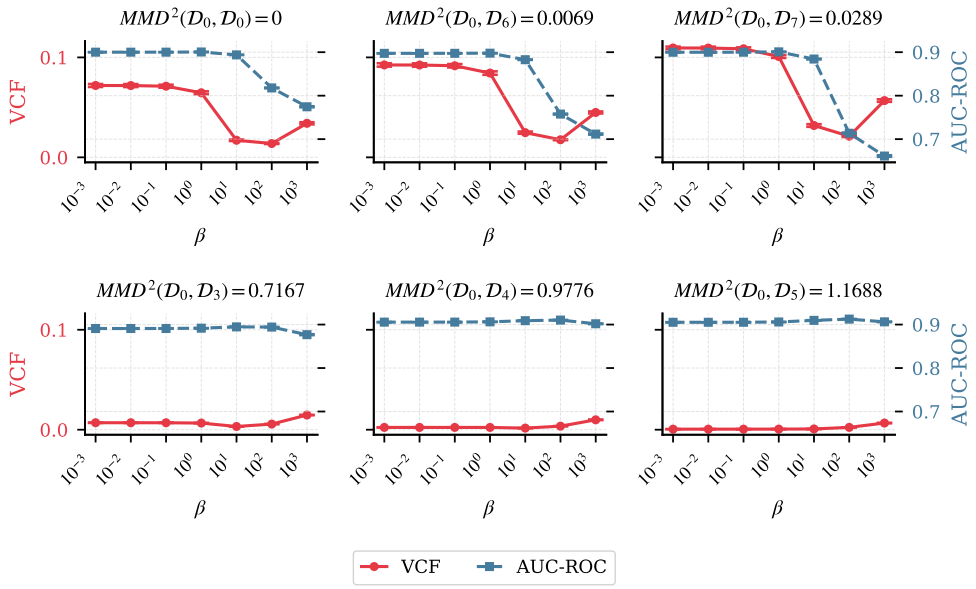


Figure 5.4: Synthetic: VCF and AUC-ROC versus β for STRATEGY 2 (warm-up then CIP) across the in-domain reference and shifted domains ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

5.1.2 Strategy Comparison: Joint vs Two-Phase

Comparing STRATEGY 1 (Figs. 5.1, 5.3) with STRATEGY 2 (Figs. 5.2, 5.4), the two strategies are indistinguishable at lower β on synthetic data. At $\beta = 1000$, STRATEGY 2 outperforms STRATEGY 1 in terms of VCF–performance trade-off. We achieve the same value of VCF but with higher AUC-ROC. However, STRATEGY 1 exhibits more consistent HSCIC across OOD domains at high β: its HSCIC continues decreasing while STRATEGY 2's HSCIC begins increasing across domains. This is because STRATEGY 2 first trains the model to predict well on the full input features for a fixed number of epochs before applying the counterfactual invariance constraint. As a result, for STRATEGY 2, the counterfactual invariance property tends to hold only at low MMD^2 distances from the training distribution. Results for all tasks, CIP feature types, and training domains are provided in Appendix A.

5.2 Real Data

On the ACS income task the true SCM is unknown, so VCF cannot be computed and HSCIC is the only invariance certificate, estimated from observational data. We report results for the arguably causal CIP feature type, training on Michigan (MI) and evaluating on the remaining states. Results for all tasks, CIP feature types, and training domains (California, Texas, Michigan) are provided in Appendix A.

5.2.1 Feature Deletion Does Not Block Proxies

Removing the counterfactual variables from the input (BASELINE 1) does not eliminate counterfactual sensitivity because proxy features remain in the data. As seen in Fig. 5.5, BASELINE 1 achieves lower in-domain AUC than BASELINE 0 (informative signal is discarded), yet the CIP strategies at low-to-moderate β match or exceed BASELINE 0 in-domain AUC while achieving lower HSCIC. CIP is

therefore more precise than deletion: it suppresses the dependence on the counterfactual variable without discarding the signal. This finding is consistent with the observation that removing a counterfactual variable does not prevent indirect dependence through correlated proxy variables (Kusner et al., 2017).

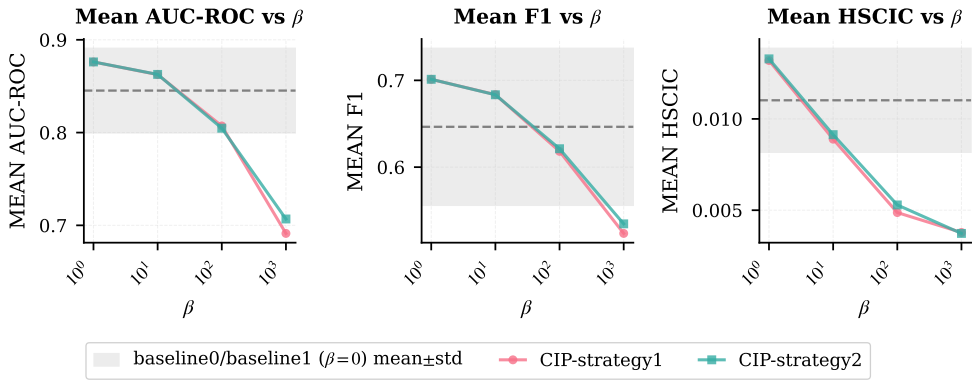


Figure 5.5: Income task, arguably causal CIP features, MI training domain: mean AUC-ROC, Macro-F1, and HSCIC versus the regularisation parameter β. The shaded band shows the unconstrained baseline mean $\pm$ std.

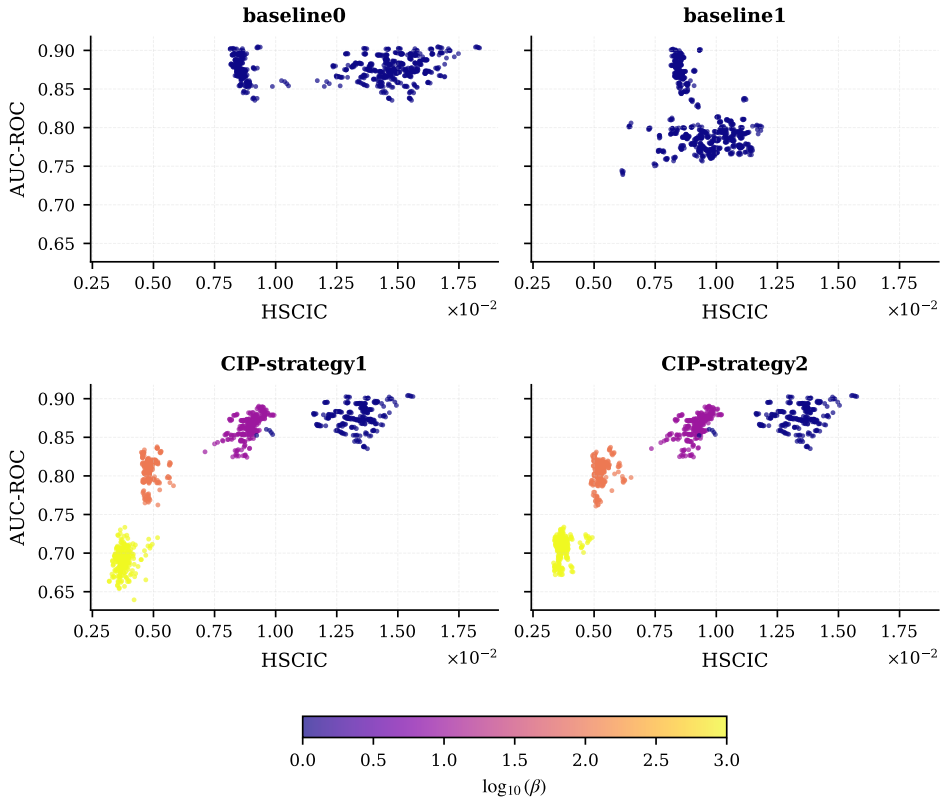


Figure 5.6: Income task, arguably causal CIP features, MI training domain: joint distribution of HSCIC and AUC-ROC across the baseline and CIP strategies.

Fig. 5.6 shows the joint distribution of (AUC-ROC, HSCIC) across states for each configuration.

The AUC-ROC distributions of Strategy 1 and Strategy 2 are similar up to $\beta = 100$. Baseline 1 achieves the lowest mean across all performance metrics, confirming that removing the counterfactual features degrades the model. Fig. 5.5 summarises the mean AUC-ROC, Macro-F1, and HSCIC as a function of β, with the unconstrained baseline band shown for reference. As in the synthetic setting, HSCIC decreases monotonically with β, while AUC-ROC and F1 remain within the baseline band up to $\beta \approx 10$ before degrading. The two CIP strategies overlap almost exactly until high β, consistent with the synthetic results.

5.2.2 MMD–Performance Relationship

Figs. 5.7 and 5.8 report the OOD AUC-ROC of each held-out state against its MMD distance from MI.

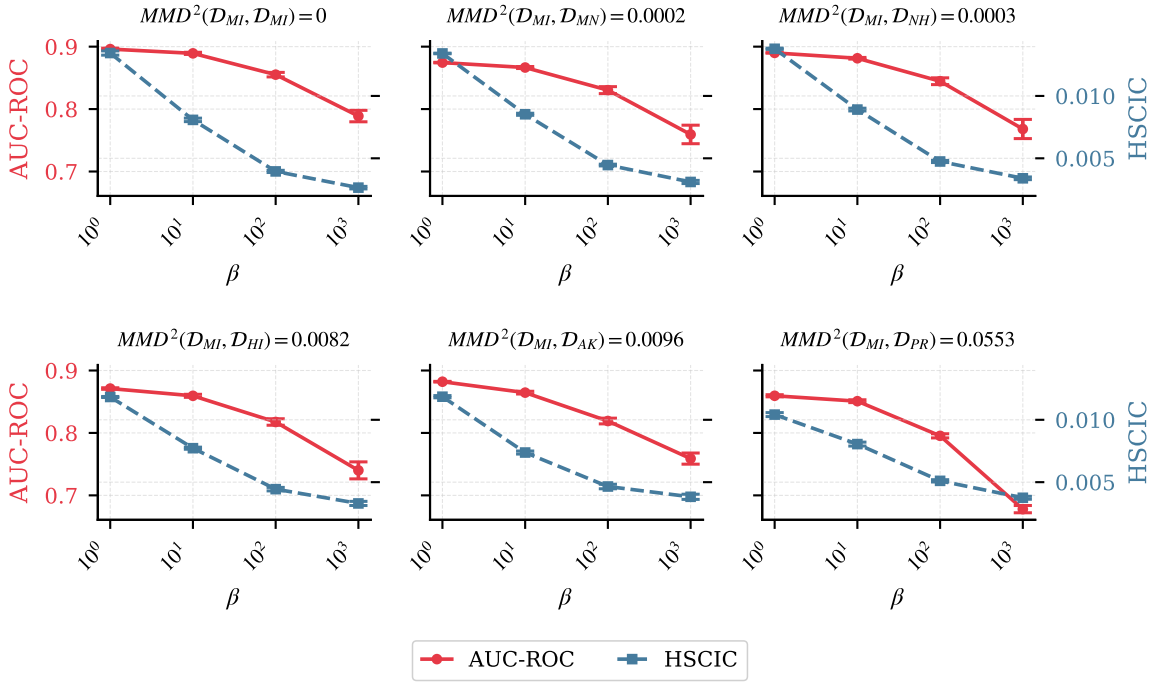


Figure 5.7: Income task, arguably causal CIP features, MI training domain: HSCIC and AUC-ROC versus β for Strategy 1 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

Unlike the synthetic setting (MMD range 0.007–1.16), the real-data MMD differences are insufficient to expose Baseline 0 fragility: all models degrade gradually together as MMD increases, and no divergence is visible between Baseline 0 and the CIP strategies at the high-MMD tail. Nevertheless, the CIP strategies maintain lower HSCIC uniformly across states, indicating that the invariance certificate holds across the observed MMD range. The synthetic experiments demonstrate the mechanism under controlled large shifts; the real data supports the certificate's consistency across states.

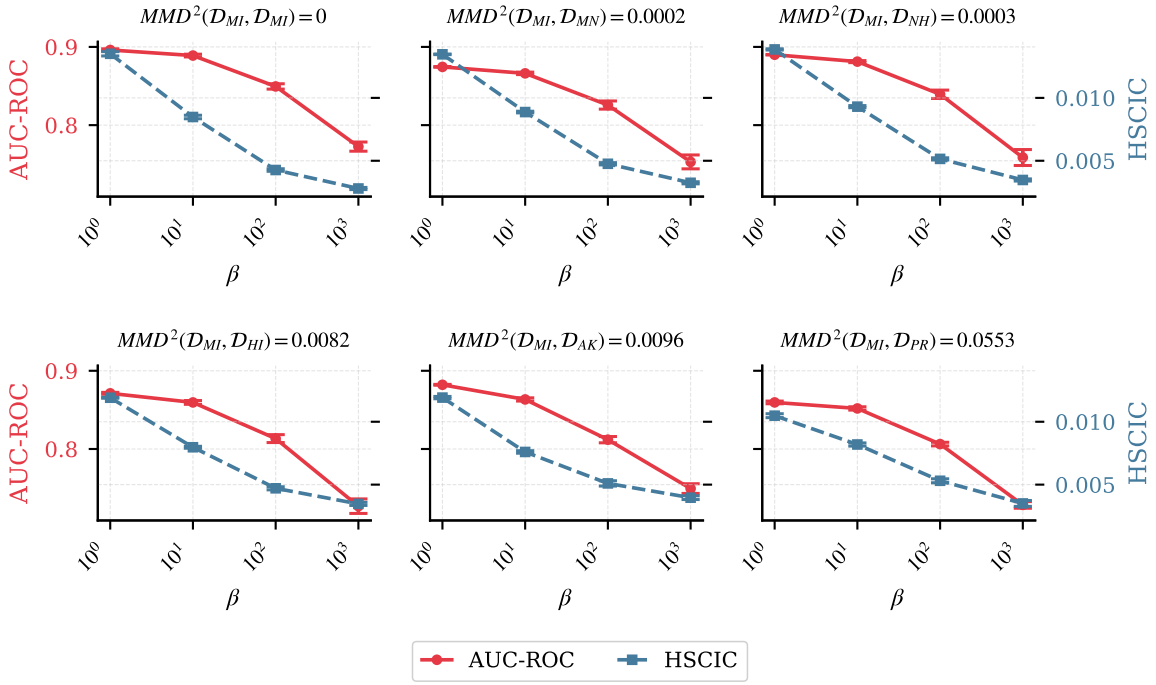


Figure 5.8: Income task, arguably causal CIP features, MI training domain: HSCIC and AUC-ROC versus β for STRATEGY 2 across the in-domain and out-of-domain ordered by MMD^2 (two domains with lowest MMD and the three with highest MMD).

5.2.3 Feature Attribution: Role of the trade-off parameter on model explanation

Figs. 5.9 and 5.10 report the pooled OOD SHAP attributions for STRATEGY 1 and STRATEGY 2 respectively as β increases, with the counterfactual (A) and abduction (W) features highlighted. The SHAP analysis confirms that the unconstrained model relies on A variables as the largest feature group: at $\beta = 1$ the counterfactual feature OCCP dominates the attribution. As β increases, A-attribution decays while the remaining features compensate; at $\beta = 1000$ the top features are no longer the counterfactual variables. A larger penalty therefore reduces the overall contribution of counterfactual variables to the prediction. However, it should be noted that model explanations are not conclusive evidence of counterfactual invariance: SHAP values only quantify how much each feature contributes to predictions without accounting for causal structure (Kaddour et al., 2022). Model explanations that do not consider causal relationships between variables must therefore be interpreted with care.

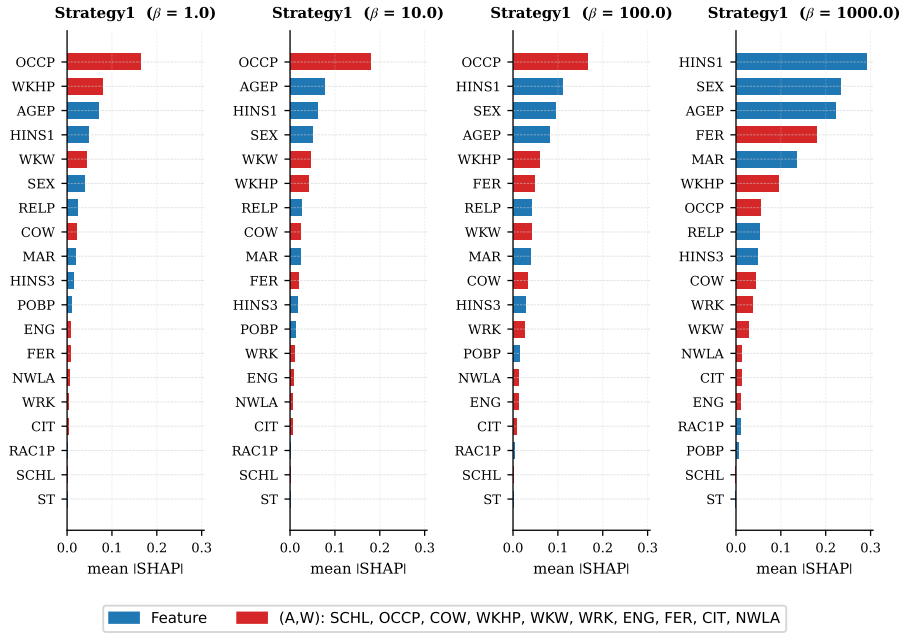


Figure 5.9: SHAP analysis: income task, arguably causal CIP features, MI training domain (OOD), STRATEGY 1 (joint optimisation). Counterfactual and Abduction (A, W) features are shown in a contrasting colour; their attribution shrinks as β grows.

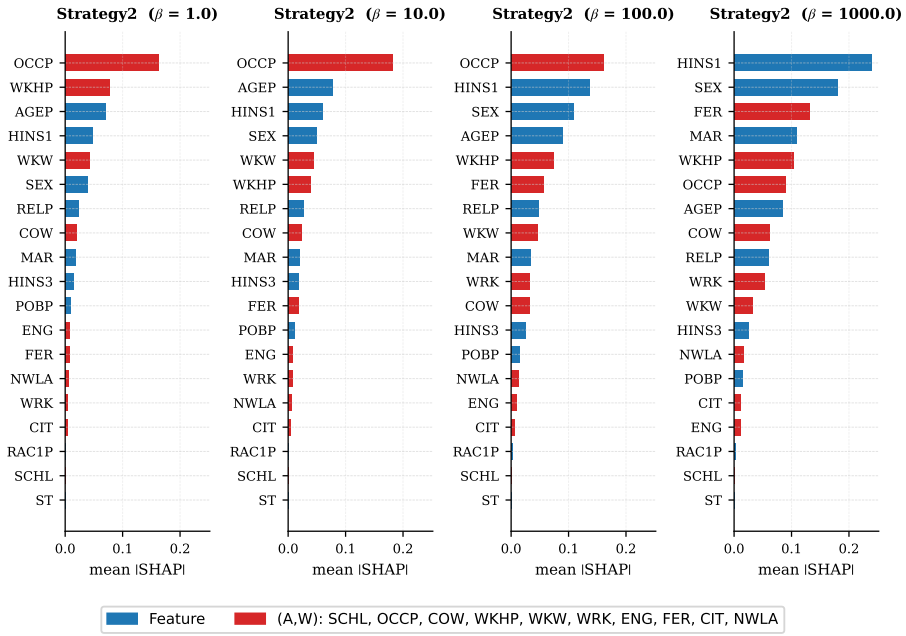


Figure 5.10: SHAP analysis: income task, arguably causal CIP features, MI training domain (OOD), STRATEGY 2 (warm-up then CIP). Counterfactual and Abduction features (A, W) are shown in a contrasting colour; their attribution shrinks as β grows.

6. Conclusion, Limitations and Future Work

6.1 Summary of Findings

The Counterfactual Invariance Prediction (CIP) framework provably enforces counterfactual invariance given certain assumptions. The main findings of our experiments are as follows.

HSCIC as an invariance certificate. The Hilbert–Schmidt Conditional Independence Criterion (HSCIC) can serve as a certificate for learned invariance. The HSCIC value decreases monotonically with the trade-off parameter β, independently of classifier accuracy metrics.

Learnable without performance collapse. For low-to-moderate regularisation strength ($\beta \leq 10$), all CIP methods converge to achieve 30%–40% reductions in HSCIC on both synthetic and real data while staying within 1%–2% AUC-ROC of the unconstrained $\textsc{Baseline}\ 0$.

Strategy matters at extreme regularisation. $\textsc{Strategy}\ 1$ (joint optimisation) and $\textsc{Strategy}\ 2$ (warm-start) coincide up to $\beta = 10$ but diverge above $\beta \geq 100$, where $\textsc{Strategy}\ 2$ significantly outperforms $\textsc{Strategy}\ 1$ in preserving AUC. However, across domains, $\textsc{Strategy}\ 1$ maintains more consistent relation between VCF and HSCIC as β grows.

Feature deletion is insufficient. Simply deleting sensitive attributes from the input fails to guarantee invariance, as indirect proxy pathways remain accessible. CIP suppresses these pathways by retaining all input features and penalising increases to HSCIC, achieving learnable invariance that feature deletion cannot.

Counterfactual invariance in A transfers under domain shift without retraining. Across all held-out target domains, CIP-trained models maintain notably lower HSCIC than the unconstrained baseline $\textsc{Baseline}\ 0$, suggesting the learned invariance is not merely an in-distribution artefact. Once trained to be counterfactually invariant in A, models appear to retain this property under distributional shift, provided the shift does not violate the causal structure assumed during training.

6.2 Limitations

Finite-Sample HSCIC Approximation. The HSCIC criterion is estimated empirically from kernel matrices, which introduces instability at extreme β values. In this regime, VCF and HSCIC diverge: HSCIC continues to decrease while VCF rebounds, suggesting the finite-sample estimator deteriorates under severe regularisation. As a result, HSCIC alone is not a reliable selection criterion at high β, and practitioners should monitor both HSCIC and VCF together.

Real-Data Distribution Constraints. The income task on US states involves relatively modest inter-state distributional shifts. Because the domains cluster closely together, the fragility of A-reliance in unconstrained models does not translate into measurable AUC-ROC differences. Fully validating the central claim that $\textsc{Baseline}\ 0$ degrades faster than CIP under hard OOD conditions would require either (a) a real-world dataset with larger natural distributional shifts across domains, or (b) synthetic experiments, as provided here, where shift magnitude can be directly controlled.

Structural Causal Model Assumption. The CIP framework requires a known causal structure to specify the counterfactual variables A and their proxy confounders W, an assumption that is often difficult to satisfy in practice. The sensitivity of CIP to feature-type misspecification remains an open

question. This reflects a broader challenge in causal machine learning, where the causal structure must either be assumed or inferred from data (Spirtes et al., 2000).

6.3 Future Research Directions

Comparison with Causal Discovery. This work assumes causal structure is known. Integrating causal discovery algorithms to learn A, W, and Z directly from data would remove this assumption. Comparing CIP-induced invariance against structure-induced invariance from discovered DAGs would shed light on the trade-off between assumption strength and practical performance. Recent work has explored learning invariant representations from observational data without explicit causal discovery, using independence and robustness criteria.

Extension to Higher-Dimensional Data. The current framework operates on tabular data with moderate feature dimensionality. Extending it to high-dimensional inputs such as images, text embeddings, or structured data is a natural future direction. The primary obstacle is the computational cost of the HSCIC estimator: computing $\widehat{\mathrm{HSCIC}}$ requires forming and inverting the kernel matrix $\mathbf{K}_z \in \mathbb{R}^{n\times n}$ and storing the Gram matrices $\mathbf{K}_x, \mathbf{K}_y \in \mathbb{R}^{n\times n}$ (Eq. (3.2.5)).

Generate Counterfactuals from Observational Data. In this work, counterfactuals are generated only in the simulation setting, where the SCM is fully specified and noise terms can be recovered. Learning counterfactual generation directly from observational data using deep generative models would remove the requirement for a known SCM while preserving the identifiability conditions of Thm. 3.3.2.

Theoretical Guarantees on Counterfactual Stability. A theoretical characterisation of when HSCIC minimisation guarantees counterfactual prediction stability, specifying the required causal assumptions and sample complexity would provide more principled guidance for practitioners. The current analysis is empirical; asymptotic guarantees on the relationship between HSCIC and VCF, drawing on kernel-based concentration inequalities (Gretton et al., 2012), would strengthen the theoretical foundations.

Systematic Kernel Selection. Throughout this work, we use the Radial Basis Function kernel. A systematic comparison of how kernel choice affects counterfactual invariance and model performance remains an open direction.

Acknowledgements

I want to acknowledge AIMS and its funders for supporting this work. I also extend my gratitude to Google DeepMind for their support through the Google DeepMind Scholarship and by providing the computation resources, as well as my supervisor, Dr. Krikamol Muandet from CISPA Helmholtz Center for Information Security, Germany, for his invaluable guidance.

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